AN AI ECOSYSTEM FOR FAKE JOB SCAM DETECTION AND FRAUD-AWARE JOB RECOMMENDATION

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October 2026

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INDIVIDUAL THESIS

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Individual component: Employer Legitimacy Verification

Group ID: R26-SE-002

Report submitted in partial fulfilment of the requirements for the

BSc (Hons) degree in Information Technology

Specializing in Software Engineering

Department of Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Online employment fraud is a growing threat in developing economies, where job seekers are increasingly targeted through deceptive advertisements on local platforms. In Sri Lanka, fake job posts are disseminated in both English and Sinhala across platforms such as ikman.lk, exploiting users who lack tools to verify job authenticity. This research presents a multilingual fake job post detection system that classifies job advertisements as Legitimate, Suspicious, or Fake using fine-tuned XLM-RoBERTa, a cross-lingual transformer model trained on 100 languages. A novel dataset of 1,724 Sri Lankan job posts was collected through automated web scraping of ikman.lk and optical character recognition of poster images from topjobs.lk using Claude Vision API. This dataset was merged with 2,866 posts from the publicly available EMSCAD dataset to form a combined training corpus of 4,590 posts. Class imbalance was addressed through inverse frequency weighting applied during cross-entropy loss computation. The fine-tuned model achieved a test accuracy of 88.5%, ROC-AUC of 0.9239, and weighted F1-score of 0.91. A three-tier classification framework combining model probability outputs with a rule-based signal scoring system of 17 linguistic and contextual indicators was developed to improve precision in ambiguous cases. Explainability was incorporated through LIME (Local Interpretable Model-Agnostic Explanations) and leave-one-out (LOO) token attribution, providing human-readable justifications for each prediction. The system was deployed as a FastAPI backend integrated with a React Native mobile application, enabling real-time scanning of job poster images. This work demonstrates that multilingual transfer learning is effective for low-resource language contexts and contributes a reusable Sri Lankan job post dataset and annotation framework for future research

Keywords: fake job detection, XLM-RoBERTa, multilingual NLP, transfer learning, explainable AI

ACKNOWLEDGEMENTS

I wish to express my sincere gratitude to my research supervisor for the continuous guidance and constructive feedback provided throughout the duration of this project. I am thankful to the FraudAware group members whose collaborative effort on the overall system architecture enabled the integration of individual components into a cohesive product. I also acknowledge the use of the EMSCAD dataset made publicly available on Kaggle by researchers studying employment fraud. The Claude Vision API provided by Anthropic was used for optical character recognition of job poster images during dataset construction. This research was conducted independently without external funding.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
AUC	Area Under the Curve
BERT	Bidirectional Encoder Representations from Transformers
CPU	Central Processing Unit
CSV	Comma-Separated Values
DSR	Design Science Research
EMSCAD	Employment Scam Aegean Dataset
EPF / ETF	Employees' Provident Fund / Employees' Trust Fund
FN	False Negative
FP	False Positive
FR	Functional Requirement
GPU	Graphics Processing Unit
HTTP / HTTPS	Hypertext Transfer Protocol / Secure
ICITR	International Conference on Information Technology Research
JSON	JavaScript Object Notation
LIME	Local Interpretable Model-Agnostic Explanations
LOO	Leave-One-Out
ML	Machine Learning
NFR	Non-Functional Requirement
NLP	Natural Language Processing
OCR	Optical Character Recognition
PII	Personally Identifiable Information
REST	Representational State Transfer
ROC	Receiver Operating Characteristic

Abbreviation	Meaning
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
TN	True Negative
TP	True Positive
URL	Uniform Resource Locator
XLM-RoBERTa	Cross-lingual Language Model based on RoBERTa

1 INTRODUCTION

1.1 Overall project context

FraudAware is a group research project developed as part of the final-year capstone at the Sri Lanka Institute of Information Technology. The project addresses the broader problem of digital fraud in Sri Lanka's online ecosystem, spanning multiple verticals including fake job advertisements, fraudulent e-commerce listings, and financial scam detection. The overall system integrates five individual detection components into a unified mobile application that allows users to scan, verify, and report suspicious digital content encountered in their daily online activity. The project targets Sri Lankan internet users who are increasingly exposed to online fraud through local platforms and social media. The group solution architecture consists of a React Native mobile frontend, individual FastAPI microservice backends for each detection component, and a shared notification and reporting layer.

1.2 Individual component context

This individual component addresses the detection of fraudulent job advertisements on Sri Lankan online platforms. Fake job posts lure victims through promises of high salaries and flexible work arrangements, then extract registration fees, training deposits, or personal data from applicants. This component accepts a job advertisement image from the FraudAware mobile application and returns a three-tier classification result — Legitimate, Suspicious, or Fake — together with a word-level explanation of the indicators that informed the decision.

The component is self-contained with its own data collection pipeline, machine learning model, and REST API backend. It exposes a /predict endpoint that is consumed by the shared FraudAware mobile frontend. Text extraction from job poster images is handled internally using the Claude Vision API, meaning the mobile app only needs to send an image and receives a complete classification response in return.

1.3 Problem statement

Sri Lankan job seekers are exposed to a significant volume of fraudulent employment advertisements, particularly on platforms such as ikman.lk and through WhatsApp-shared job

poster images. These posts typically promise unrealistically high salaries, require no prior experience, and request upfront payments framed as registration fees or training deposits. Victims lose money and, in some cases, personal documents submitted during a fake application process.

Existing job platforms in Sri Lanka do not provide any automated fraud detection capability. Users rely entirely on their own judgment when evaluating job offers, which is especially problematic for first-time job seekers and low-income workers who lack the experience to identify linguistic and structural indicators of fraud. Furthermore, a significant portion of fraudulent posts in Sri Lanka are written in Sinhala, yet no detection tool exists that covers Sinhala-language content. This leaves a large segment of the population with no technological means to verify the authenticity of a job advertisement before responding to it.

1.4 Research gap

Prior work on fake job detection has focused primarily on English-language datasets, most notably the EMSCAD dataset, which contains 17,880 job posts sourced from the United States. Studies using this dataset have applied machine learning approaches including BERT, logistic regression, and random forest classifiers, achieving high accuracy figures. However, none of these studies address the multilingual context of Sri Lanka, where job advertisements appear in both English and Sinhala with distinct linguistic patterns, salary conventions, and fraud tactics specific to the local market.

Beyond the language gap, existing systems produce only binary classifications (fake or legitimate) and do not provide any explanation of why a post was flagged. This limits their practical utility for non-technical users who need to understand the reason behind a warning in order to act on it. There is also no publicly available labelled dataset of Sri Lankan job posts, which restricts future research in this domain and means that any model trained purely on foreign data will not generalize well to local platform content.

1.5 Aim and objectives

The aim of this research is to design, develop, and evaluate a multilingual fake job post detection system capable of classifying English and Sinhala job advertisements as Legitimate, Suspicious, or Fake, and providing explainable justifications for each classification result.

The objectives are:

1. To collect and construct a labelled dataset of Sri Lankan job advertisements in English and Sinhala from ikman.lk and topjobs.lk.
2. To fine-tune a multilingual transformer model (XLM-RoBERTa) on a combined dataset of Sri Lankan posts and the publicly available EMSCAD dataset.
3. To implement a three-tier classification framework that combines model output probabilities with a rule-based signal scoring system to handle ambiguous cases.
4. To integrate LIME and leave-one-out (LOO) token attribution to produce per-prediction word-level explanations that are meaningful to non-technical users.
5. To deploy the detection system as a FastAPI microservice integrated with the FraudAware React Native mobile application.

1.6 Research questions or hypotheses

This research is guided by the following three research questions:

RQ1: Can a fine-tuned XLM-RoBERTa model achieve competitive fake job detection performance on a combined Sri Lankan and EMSCAD dataset compared to reported benchmarks on English-only datasets?

RQ2: Does the addition of a rule-based signal scoring system improve classification precision on ambiguous cases where model probability alone is insufficient for a confident prediction?

RQ3: Can LIME and token-level attribution produce meaningful word-level explanations for fake job post predictions that are interpretable by non-technical Sri Lankan users?

1.7 Scope limitations and assumptions

This component covers English and Sinhala language job posts only. Tamil posts were identified during data collection but excluded from model training due to insufficient volume (only 8 posts collected), which was not enough to train a reliable Tamil classifier.

The system classifies text extracted from job post images using OCR. Image quality directly affects extraction accuracy — blurry or low-resolution poster images may result in incomplete text and reduced prediction reliability.

The rule-based signal scoring system was developed based on patterns observed specifically in the Sri Lankan job market, such as EPF/ETF benefit mentions as legitimacy indicators and WhatsApp-only contact as a fraud indicator. These rules may not generalize to job markets in other countries without modification.

The model was trained on a combined dataset of 4,590 posts. Performance is expected to improve with a larger and more balanced dataset, particularly for Sinhala-language posts where only 308 training samples were available.

The system assumes that the FastAPI backend server is reachable from the mobile device, either over the same local network or via a tunnelling service, and that the mobile device has permission to access the camera or photo gallery.

1.8 Individual contribution

The specific contributions made by this individual component to the FraudAware group project are as follows:

1. **Dataset construction** — A novel labelled dataset of 1,724 Sri Lankan job posts was collected from ikman.lk (500 posts via web scraping) and topjobs.lk (1,224 posts via OCR of poster images using the Claude Vision API), with Unicode-based language detection to correctly identify Sinhala content.
2. **Annotation framework** — A set of annotation guidelines was developed defining fake indicators (upfront fee requests, vague descriptions, personal Gmail/Yahoo contacts, unrealistic salary promises) and legitimate indicators (Pvt Ltd/PLC company names, corporate email addresses, EPF/ETF mentions), used to guide both automated and manual labelling.
3. **Multilingual detection model** — XLM-RoBERTa was fine-tuned on the combined Sri Lankan and EMSCAD dataset, producing a model capable of classifying both English and Sinhala job posts.
4. **Three-tier classification framework** — A signal scoring system of 17 linguistic and contextual rules was implemented and combined with model output probabilities to produce Legitimate, Suspicious, or Fake classifications.

5. **Explainability integration** — LIME and token-level leave-one-out (LOO) attribution was integrated into the prediction pipeline to generate per-prediction word-level justifications displayed in the mobile app.
6. **Backend and mobile integration** — A FastAPI microservice was developed and integrated with two React Native screens (JobPostScreen and JobPostResultScreen) within the shared FraudAware mobile application.

1.9 Thesis structure

The remainder of this thesis is structured as follows:

Chapter 2 presents the literature review, covering prior work on fake job detection, the theoretical foundations of multilingual transformer models, and existing systems and datasets relevant to this research. It concludes with a synthesis of identified gaps that this work addresses.

Chapter 3 describes the research methodology, including the data collection pipeline, dataset construction and labelling process, model training procedure, and evaluation design.

Chapter 4 presents the product and solution design, covering system requirements, component architecture within the FraudAware platform, design alternatives considered, and the detailed design of the classification and explainability pipeline.

Chapter 5 covers implementation, integration, and testing, describing the development environment, how each part of the system was built, how the component integrates with the mobile frontend, and the testing performed.

Chapter 6 reports the quantitative results from model evaluation and validates the three-tier classification framework and explainability outputs.

Chapter 7 discusses the findings, compares results against literature baselines, and identifies limitations and implications of the work.

Chapter 8 concludes the thesis, summarizes the achievement of objectives, and outlines directions for future work.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review was conducted through a structured search of Google Scholar, IEEE Xplore, and ACM Digital Library. The primary search terms used were: *fake job detection*, *fraudulent employment NLP*, *multilingual text classification*, *XLM-RoBERTa fine-tuning*, *transfer learning for fraud detection*, and *LIME LOO text explainability*. The review was limited to work published between 2018 and 2025 to ensure relevance to contemporary deep learning methods.

Studies were included if they addressed one or more of the following: automated fraud detection in text, multilingual natural language processing, transformer model fine-tuning for classification tasks, or explainable AI applied to text classifiers. Studies using the EMSCAD dataset were specifically sought to establish a performance baseline for comparison. Work on Sri Lankan NLP was also reviewed, though the literature in this area remains limited.

2.2 Domain and theoretical foundations

Fake job post detection falls within the broader domain of fraud detection in online platforms, which has been studied using rule-based systems, classical machine learning, and more recently deep learning approaches. The core challenge is that fraudulent posts are deliberately written to appear legitimate, requiring models that can capture subtle semantic patterns rather than simple keyword matches.

2.2.1 Transfer learning and transformer models

The theoretical foundation of this work rests on transfer learning — the practice of using a model pre-trained on large corpora and fine-tuning it on a smaller task-specific labelled dataset. XLM-RoBERTa [1], developed by Conneau et al. at Facebook AI Research, is a transformer-based model pre-trained on 100 languages using masked language modelling on 2.5 terabytes of filtered CommonCrawl data. Unlike multilingual BERT, XLM-RoBERTa uses a larger vocabulary (250,000 tokens) and was trained with more data per language, resulting in stronger cross-lingual representations. This makes it particularly suitable for Sri Lanka's bilingual English-Sinhala context, where no large Sinhala-specific language model currently exists.

2.2.2 Explainability methods

LIME (Local Interpretable Model-Agnostic Explanations) [2], introduced by Ribeiro et al., provides post-hoc interpretability by approximating the behavior of a complex model locally around a specific input. For text classification, LIME perturbs the input by randomly removing words and observing how the model's output probability changes, then fits a simple linear model to identify which words contributed most to the prediction. LOO (SHapley Additive explanations) [3], introduced by Lundberg and Lee, assigns each feature a Shapley value derived from cooperative game theory, representing its average marginal contribution across all possible subsets of features. Both methods are model-agnostic and can be applied to any classifier, including transformer models.

2.3 Related research

Several studies have applied machine learning to the EMSCAD dataset for fake job detection. Mahbub et al. [4] compared BERT, XGBoost, and LSTM classifiers and reported an F1-score of 0.98 for the fake class. However, this result must be interpreted cautiously — the dataset is severely imbalanced with only 866 fake posts against 17,014 legitimate posts, and high accuracy figures can be achieved by a model that simply predicts legitimate for most inputs. Amaar et al. [5] applied a broader set of classifiers including SVM, random forest, and ensemble methods on EMSCAD, reporting up to 97.8% accuracy, and found that ensemble methods consistently outperformed single classifiers. Neither study addressed multilingual content or provided any explanation of individual predictions.

In the domain of multilingual fraud detection, Aziz et al. [6] applied multilingual BERT to phishing URL detection across several languages and demonstrated that cross-lingual models can transfer knowledge effectively even for low-resource languages. This provides supporting evidence for the use of XLM-RoBERTa in the Sri Lankan English-Sinhala context. Kumari and Singh [7] studied social media fraud detection and showed that augmenting neural model outputs with rule-based feature signals improved precision on borderline cases — a finding that directly informed the signal scoring system developed in this work.

Research specifically targeting Sri Lankan NLP is sparse. Most available Sinhala NLP work focuses on sentiment analysis and machine translation rather than fraud detection, and no prior study has constructed a labelled Sri Lankan job post dataset or applied a multilingual model to local employment fraud.

2.4 Existing systems products and techniques

No commercially deployed fake job detection system specifically targets Sri Lanka or the Sinhala language. The following existing systems and techniques were reviewed to understand the current landscape:

Platform-level moderation — ikman.lk and other local job platforms rely on user reporting and manual moderation to handle fraudulent posts. There is no automated detection layer visible to users, and fraudulent posts often remain live for days before being removed.

LinkedIn fraud detection — LinkedIn applies proprietary machine learning models to detect fake job posts at the platform level. However, this system is internal, not accessible via API, and does not cover local Sri Lankan platforms where the majority of fraud occurs.

EMSCAD baseline tools — The authors of the EMSCAD dataset published a baseline Jupyter notebook applying logistic regression and SVM to the dataset. These tools are English-only, produce binary outputs with no explanation, and are not integrated into any user-facing application.

General scam detection apps — Applications such as Truecaller (for phone number verification) and various browser-based phishing detectors address adjacent fraud problems but do not classify job post text and are not designed for the Sri Lankan market.

OCR and vision tools — Google Cloud Vision and Tesseract OCR have been used in related work to extract text from images. In this research, the Claude Vision API was selected over Tesseract due to its superior ability to handle mixed-language poster images with varied fonts and layouts common in Sri Lankan job advertisements.

2.5 Gap synthesis and conceptual framework

The review of related work reveals three clear gaps that this research directly addresses:

Gap 1 — Language coverage: All existing fake job detection research uses English-only datasets. No system exists that covers Sinhala, leaving the majority of Sri Lankan job seekers who encounter Sinhala-language fraudulent posts without any technological protection.

Gap 2 — Explainability: Existing systems produce binary classification labels without any explanation of the indicators that led to the decision. For a non-technical user deciding whether to respond to a job advertisement, knowing that a post was flagged because it contains a registration fee request and a personal Gmail contact is far more actionable than a label alone.

Gap 3 — Sri Lankan dataset: There is no publicly available labelled dataset of Sri Lankan job posts. Models trained solely on the EMSCAD dataset (US-sourced) do not capture the linguistic patterns, salary conventions, and fraud tactics specific to the Sri Lankan market, limiting their practical utility locally.

The conceptual framework of this research positions XLM-RoBERTa fine-tuning as the core detection mechanism, selected for its proven cross-lingual capability across 100 languages, including Sinhala. This is augmented by a rule-based signal scoring system that captures Sri Lanka-specific fraud indicators not well represented in the EMSCAD training data. LIME and token-level attribution provide the explainability layer, and the whole pipeline is delivered through a mobile-first interface suited to Sri Lanka's smartphone-dominant internet usage patterns.

3 RESEARCH METHODOLOGY

This chapter presents the methodology adopted to design, develop, and evaluate a multilingual fake job post detection system tailored to the Sri Lankan employment context. The research follows a Design Science Research (DSR) approach, in which the primary output is a functional artefact — a fine-tuned XLM-RoBERTa classifier integrated with a rule-based signal scoring layer and a mobile-facing REST API — rather than a purely theoretical contribution. Quantitative evaluation methods are used to measure the performance of the artefact against held-out test data. The chapter is organised into seven sections covering research design, study materials, experimental procedure, evaluation metrics, data analysis, ethical and responsible AI considerations, and reproducibility.

3.1 Research design

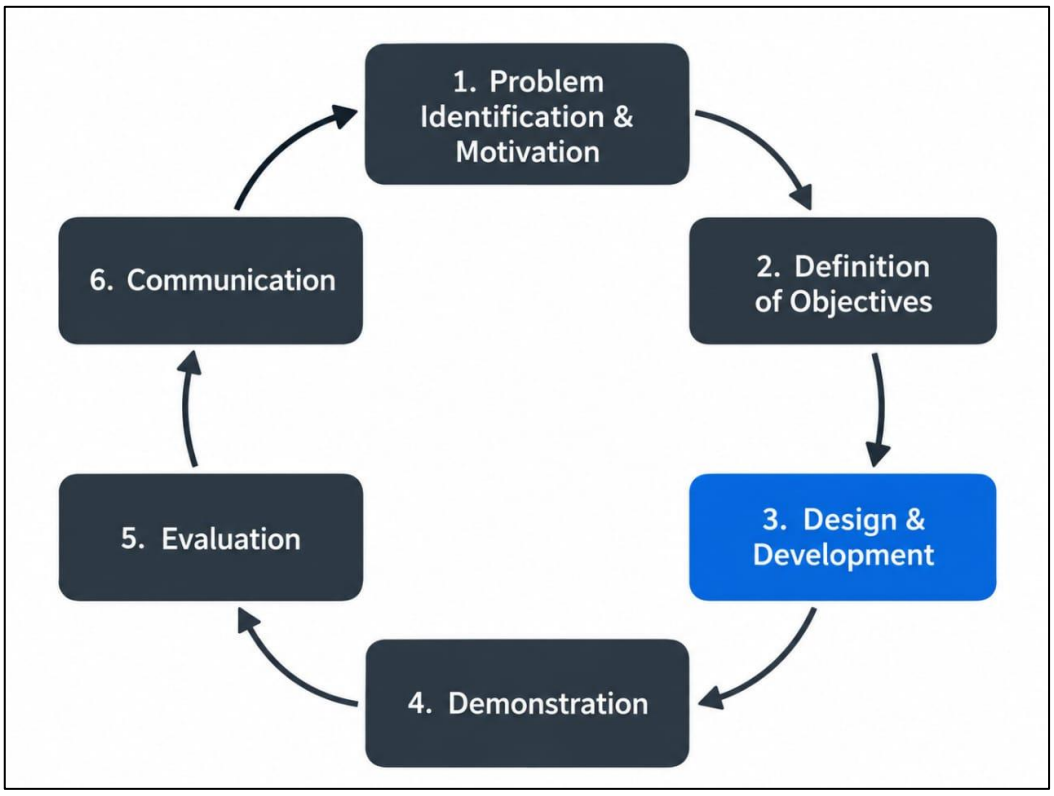


Figure 3.1: DSR Methodology Phases

This research adopts a positivist paradigm, treating fake job post detection as an empirical problem that can be studied through systematic data collection, model training, and quantitative

measurement of outcomes. The study is applied and experimental in nature: it applies established machine learning techniques to a domain-specific problem — multilingual employment fraud in the Sri Lankan context — and evaluates the resulting system against held-out test data.

The research follows a Design Science Research (DSR) methodology. DSR is appropriate here because the goal is to solve a practical problem (protecting job seekers from fraudulent postings) while contributing knowledge about how multilingual transformer models can be adapted to low-resource, code-mixed language environments. The primary output is a functional artefact rather than a theoretical model.

A quantitative evaluation strategy is employed, using standard classification metrics (accuracy, F1-score, ROC-AUC, confusion matrix) to assess model performance. The dataset construction phase incorporates both secondary data — the EMSCAD benchmark dataset sourced from Kaggle — and primary data collected through web scraping and OCR-based extraction from Sri Lankan job platforms, providing ecological validity that a purely benchmark-driven study would lack.

3.2 Data participants or study materials

This study does not involve human participants. The study materials consist entirely of job post text data drawn from two sources: an international benchmark dataset and locally collected Sri Lankan job postings.

3.2.1 EMSCAD Benchmark Dataset

The Employment Scam Archetypes Dataset (EMSCAD), publicly available on Kaggle, contains 17,880 job postings labelled as legitimate or fraudulent. To maintain class balance and computational feasibility, a stratified sample of 2,866 records was drawn: 2,000 legitimate posts and 866 fake posts. Stratified sampling ensured that the proportion of fraudulent records was preserved in the sample, providing sufficient minority-class examples for effective model training.

3.2.2 Sri Lankan Local Dataset

Two Sri Lankan job platforms were targeted to capture the linguistic and structural characteristics of local postings:

ikman.lk — 500 job posts were collected via automated web scraping. These posts are predominantly in English but frequently contain Tamil and Sinhala words embedded within job descriptions and contact details.

topjobs.lk — 1,224 job posts were collected as poster-style images. Since topjobs.lk renders many advertisements as image files rather than structured text, Optical Character Recognition (OCR) was performed using the Claude Vision API (claude-haiku-3 model) to extract textual content from each image.

All locally collected posts were labelled using an AI-assisted labelling pipeline. The Claude Haiku API was prompted with each post alongside a structured labelling rubric to classify each record as legitimate or fake. A random sample of 10% of AI-generated labels was manually reviewed to verify labelling consistency.

3.2.3 Combined Dataset

The EMSCAD sample and the Sri Lankan local data were merged into a single combined dataset after deduplication and format normalization. The final dataset contains 4,590 records distributed as follows:

Table 3.1: Composition of the combined dataset by source

Source	Legitimate	Fake	Total
EMSCAD (stratified sample)	2,000	866	2,866
ikman.lk (scraped)	424	76	500
topjobs.lk (OCR)	1,274	—	1,274
Combined	**3,698**	**892**	**4,590**

The combined dataset exhibits a class imbalance ratio of approximately 1:4 (fake: legitimate), which was addressed during model training through inverse-frequency class weighting.

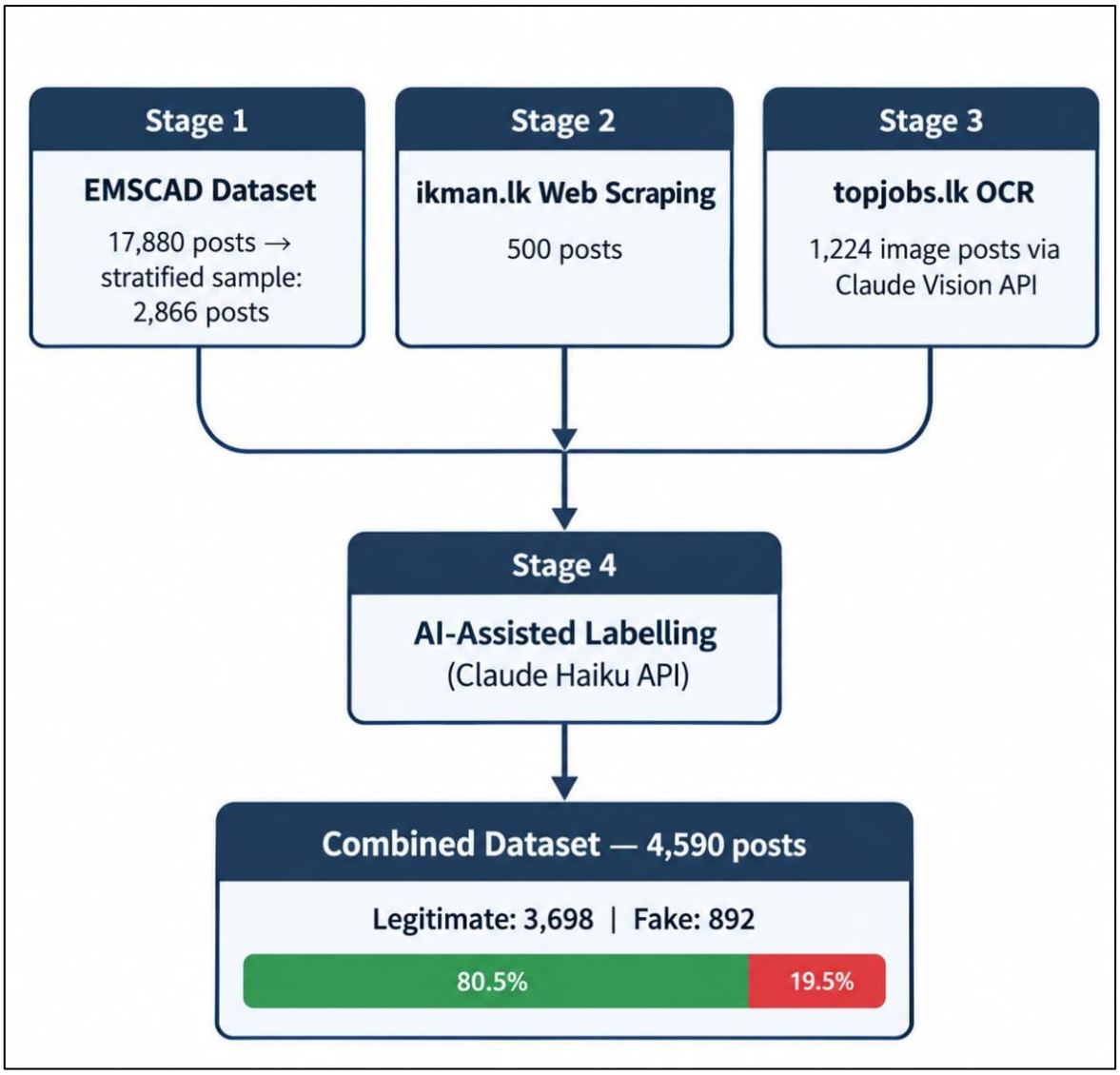


Figure 3.2: Dataset Construction Pipeline

3.3 Experimental or study procedure

The study procedure followed a sequential pipeline of six stages: data pre-processing, model architecture selection, training configuration, signal scoring, three-tier classification, and explainability integration.

3.3.1 Data Pre-processing

Each job post was subjected to a lightweight normalization pipeline prior to tokenization. This included removal of HTML tags and special characters introduced by web scraping,

normalisation of whitespace and line breaks, and lowercasing of English text. Sinhala and Tamil characters were retained without modification, as XLM-RoBERTa's multilingual vocabulary handles these scripts natively.

To identify Sri Lankan-specific posts containing Sinhala content, a character-range heuristic was applied. Posts in which more than 30% of characters fell within the Unicode range U+0D80–U+0DFF (the Sinhala script block) were flagged as Sinhala-dominant. This detection was used to adjust AI-assisted labelling prompts to include Sinhala-specific fraud indicators.

All text was tokenised using the XLM-RoBERTa tokenizer (xlm-roberta-base vocabulary, 250,002 subword tokens), which applies SentencePiece byte-pair encoding. Sequences were truncated or padded to a maximum length of 512 tokens. The dataset was partitioned using a stratified 80/20 train/test split, yielding 3,672 training records and 918 test records.

3.3.2 Model Architecture

The classification model is built on XLM-RoBERTa base (xlm-roberta-base), a multilingual transformer pre-trained on 2.5 terabytes of CommonCrawl text spanning 100 languages, including English, Sinhala, and Tamil. The model comprises 12 transformer encoder layers, 768 hidden dimensions, 12 attention heads, and approximately 278 million parameters.

A binary sequence classification head was appended to the pre-trained encoder — a single linear layer mapping the [CLS] token representation (768 dimensions) to a two-class logit output, followed by softmax activation. The entire model was fine-tuned end-to-end on the combined dataset.

3.3.3 Training Configuration

Fine-tuning was conducted using the AdamW optimiser with a learning rate of 2e-5 and weight decay of 0.01, over 5 epochs with a batch size of 16. A linear warmup schedule was applied for the first 10% of training steps, followed by linear decay to zero. Class imbalance was addressed through inverse-frequency class weighting applied to CrossEntropyLoss, computed as:

$w_c = N / (C \times n_c)$

where N is the total training samples, C is the number of classes (2), and n_c is the count of class c. This assigned a weight of approximately 2.56 to the fake class and 0.62 to the legitimate class.

3.3.4 Signal Scoring System

A complementary rule-based signal scoring system of 17 hand-crafted rules was developed to operate in parallel with the model. Each rule assigns a score increment based on the presence of a specific textual or structural feature: strong fake indicators contribute +2, moderate fake indicators +1, and legitimate indicators −1. The total signal score for each post is the sum of all applicable rule scores.

3.3.5 Three-Tier Classification

The model's fake probability (p) and the signal score (s) are combined to assign one of three risk tiers:

Fake — if ($p \geq 0.85$ AND $s \geq 2$) OR ($p \geq 0.92$)

Suspicious — if ($p \geq 0.50$) OR ($s \geq 1$), provided the Fake condition is not met

Legitimate — all remaining posts

3.3.6 Explainability Integration

Two post-hoc explainability mechanisms were integrated. LIME (LimeTextExplainer) generates 40 perturbed samples per prediction and returns the 8 highest-weight tokens driving the classification. A leave-one-out (LOO) token attribution method masks each token in turn and records the change in fake probability, identifying the tokens most responsible for the model's decision. Both mechanisms are invoked at inference time and their outputs are surfaced in the mobile application.

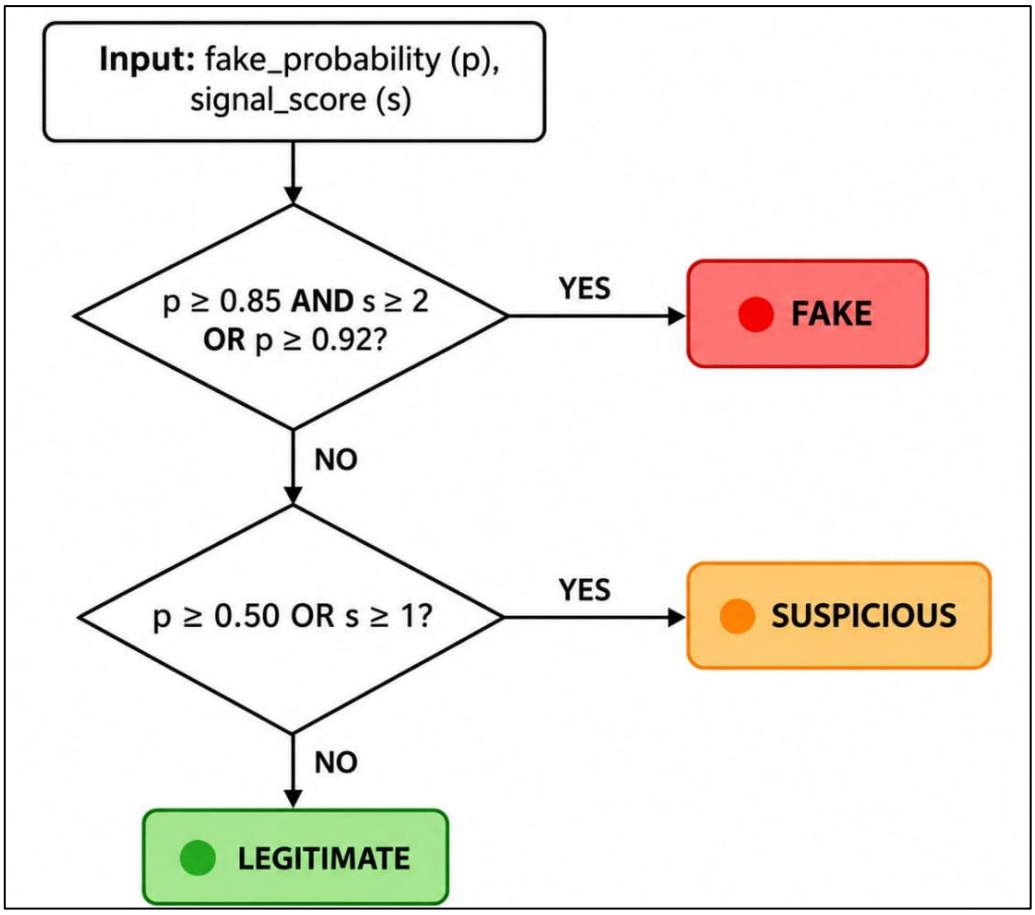


Figure 3.3: Three-Tier Classification Decision Logic

3.4 Measures and evaluation metrics

Model performance was assessed using four standard binary classification metrics, each chosen for a specific reason in the context of imbalanced fraud detection.

3.4.1 Accuracy

Accuracy measures the proportion of all predictions that are correct across both classes. It provides an overall performance indicator but can be misleading under class imbalance — a classifier that always predicts “legitimate” would achieve approximately 80% accuracy on this dataset simply by exploiting the majority class. Accuracy is therefore reported alongside the metrics below rather than used in isolation.

3.4.2 F1-Score

The F1-score is the harmonic mean of precision and recall, computed separately for the fake class and the legitimate class. In fraud detection, recall of the fake class — the ability to correctly identify fraudulent posts — is particularly important, as missing a fake post (a false negative) carries a higher real-world cost than incorrectly flagging a legitimate post (a false positive). The weighted F1-score, weighted by class support, is also reported as a single summary statistic that accounts for class imbalance.

3.4.3 ROC-AUC

The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) measures the model's ability to discriminate between classes across all possible classification thresholds. An AUC of 1.0 indicates perfect discrimination; 0.5 indicates random chance. AUC is threshold-independent, making it useful for evaluating the underlying model quality separately from the specific probability thresholds chosen for the three-tier classification logic.

3.4.4 Confusion Matrix

The confusion matrix provides a breakdown of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) on the held-out test set. This allows direct inspection of the trade-off between false positives (legitimate posts incorrectly flagged as fake) and false negatives (fake posts missed by the system), which is the central tension in any fraud detection deployment.

3.4.5 Classification Report

A full per-class classification report — including precision, recall, and F1-score for both the legitimate and fake classes — was generated on the 918-record test set to provide a complete picture of model behavior across both classes.

3.5 Data analysis

Data analysis was conducted at two levels: dataset-level analysis prior to model training, and model-level analysis after evaluation on the held-out test set.

3.5.1 Dataset-Level Analysis

Prior to training, the combined dataset was analyzed to understand its composition and potential sources of bias. The class distribution was examined to quantify the imbalance between legitimate and fake posts (3,698 vs. 892 records). The language distribution across posts was assessed using the Sinhala Unicode heuristic described in Section 3.3.1, confirming that the majority of local posts were English-dominant with code-mixed Tamil or Sinhala elements. Post length distributions (in tokens) were analyzed to determine an appropriate maximum sequence length, with the 95th percentile falling below 512 tokens, justifying the chosen truncation threshold.

3.5.2 Training Dynamics Analysis

Training and validation loss curves were monitored across all five epochs to detect overfitting or underfitting. Training accuracy reached 96.52% by the final epoch, while test accuracy stabilized at 88.5%, indicating a moderate generalization gap attributable to the domain shift between EMSCAD (international, English-only) and Sri Lankan (multilingual, locally structured) posts. No early stopping was applied, as the loss curves showed consistent improvement without significant divergence between training and validation loss.

3.5.3 Model Performance Analysis

On the 918-record held-out test set, the model achieved the following results

Table 3.2: Model performance on the held-out test set

Metric	Fake Class	Legitimate Class	Weighted
Precision	0.75	0.93	—
Recall	0.79	0.92	—
F1-Score	0.77	0.92	0.91
Accuracy	—	—	0.885
ROC-AUC	—	—	0.9239

The confusion matrix on the test set yielded 439 true legitimate (TN), 123 true fake (TP), 40 false positives (FP), and 33 false negatives (FN). The higher false positive count relative to false negatives indicates that the model errs slightly on the side of caution — flagging some legitimate posts as suspicious — which is the preferred error direction for a consumer-facing fraud detection system.

3.5.4 Explainability Analysis

LIME explanations were examined qualitatively across a sample of correctly and incorrectly classified posts. In correctly identified fake posts, the top LIME tokens consistently included phrases associated with financial requests, urgency language, and vague job descriptions. In false negatives (fake posts classified as legitimate), the LIME tokens revealed that the posts used formal language and complete contact details that resembled legitimate postings, suggesting a more sophisticated form of fraud not well represented in the training data.

3.6 Ethics privacy security and responsible AI

3.6.1 Ethical Considerations

This research does not involve human participants in any experimental capacity. No surveys, interviews, or user trials were conducted. The data used consists entirely of publicly available job postings scraped from publicly accessible platforms (ikman.lk and topjobs.lk). As such, formal ethical approval for human subjects' research was not required. However, the following ethical considerations were observed throughout the study.

3.6.2 Data Privacy

Job postings on ikman.lk and topjobs.lk are publicly visible without authentication. Nevertheless, posts that contained identifiable personal information — such as individual applicant names, personal phone numbers, or home addresses posted by private individuals rather than companies — were excluded from the dataset during the cleaning phase. The dataset was used solely for the purpose of training and evaluating the fraud detection model and was not shared or published externally.

AI-assisted labelling was performed using the Claude Haiku API. No personally identifiable information (PII) was intentionally included in API prompts. Posts containing personal contact details were anonymized (phone numbers and email addresses replaced with placeholder tokens) before being submitted to the labelling pipeline.

3.6.3 Responsible AI

The system is designed as a decision-support tool, not an automated enforcement mechanism. The three-tier classification output (Fake / Suspicious / Legitimate) is presented to the user alongside the LIME explanation tokens so that the user can inspect the reasoning and make an informed judgement. No post is automatically removed or blocked based solely on the model's output.

The model carries known limitations that users should be aware of. It was trained predominantly on English and EMSCAD-style fraud patterns; its performance on purely Sinhala or Tamil posts may be lower than reported accuracy figures suggest, as these languages are underrepresented in the training data. This limitation is disclosed in the application interface.

3.6.4 Security

The FastAPI backend does not store any job post text submitted by users. All inference is stateless and ephemeral — submitted text is processed in memory and discarded after the response is returned. The /extract-text endpoint, which accepts image uploads, validates file type and size before processing to prevent abuse. Communication between the mobile application and the backend is conducted over HTTPS to protect data in transit.

3.7 Reproducibility and research artefacts

3.7.1 Reproducibility

To ensure the reproducibility of the experimental results, the following measures were adopted throughout the study:

Random seeds were fixed at the start of all data splitting, model initialization, and LIME sampling procedures to eliminate stochasticity from results.

The stratified 80/20 train/test split was performed using a fixed random seed, ensuring that the same records appear in the training and test sets across all experimental runs.

All hyperparameters used during fine-tuning — learning rate, batch size, number of epochs, warmup proportion, and class weights — are fully documented in Section 3.3.3 and can be replicated without modification.

The LIME configuration (40 perturbed samples, 8 features, fixed random seed) is documented in Section 3.3.6 to allow identical explanation outputs to be reproduced.

The signal scoring rules (Section 3.3.4) are deterministic and require no probabilistic reproduction.

3.7.2 Research Artefacts

The following artefacts were produced as outputs of this research:

Fine-tuned model — the XLM-RoBERTa base model fine-tuned on the combined 4,590-record dataset, saved as a PyTorch model checkpoint compatible with the Hugging Face Transformers library.

Combined dataset — the merged and labelled dataset of 4,590 job posts (EMSCAD sample + Sri Lankan local data), stored in CSV format with columns for post text, source platform, language flag, and binary label.

FastAPI backend — the REST API server exposing the /predict and /extract-text endpoints, implemented in Python using FastAPI and the Hugging Face Transformers inference pipeline.

React Native mobile application — the FraudAware mobile application built with React Native Expo, comprising the JobPostScreen and JobPostResultScreen components, targeting iOS and Android.

Signal scoring module — the standalone Python module implementing the 17-rule signal scoring system, usable independently of the neural model.

ICITR 2026 conference paper — a six-page IEEE-format paper submitted to the International Conference on Information Technology Research 2026, Track 02 (Artificial Intelligence), University of Moratuwa.

All source code and configuration files are maintained in a version-controlled repository. The fine-tuned model weights and dataset are available upon request for academic purposes.

4 PRODUCT AND SOLUTION DESIGN

This chapter presents the design of the multilingual fake job post detection system developed as part of this research. It covers the functional and non-functional requirements of the system, the component architecture within the broader FraudAware ecosystem, the design alternatives considered and the decisions made, the detailed design of each component, and the standards and professional practices followed throughout the development process.

4.1 Requirements

4.1.1 Functional Requirements

The following functional requirements were identified for the fake job post detection system:

FR01 — Multilingual Text Classification

The system shall accept job post text in English, Sinhala, and Tamil and classify it as Fake, Suspicious, or Legitimate using a fine-tuned XLM-RoBERTa model combined with a rule-based signal scoring layer.

FR02 — Image-Based Job Post Extraction

The system shall accept a photograph or screenshot of a job advertisement poster and extract the textual content via OCR before passing it to the classification pipeline.

FR03 — Three-Tier Risk Output

The system shall return one of three risk tier labels — Fake, Suspicious, or Legitimate — along with a fake probability score and a signal score for each submitted job post.

FR04 — Explainability Output

The system shall return the top 8 LIME explanation tokens highlighting the words that most influenced the classification decision, displayed to the user in the mobile application.

FR05 — REST API

The system shall expose a REST API with at least two endpoints: one for text-based classification (/predict) and one for image-based text extraction (/extract-text).

FR06 — Mobile Application Interface

The system shall provide a mobile application allowing users to input job post text manually, paste copied text, or capture an image, and view the classification result with explanation.

4.1.2 Non-Functional Requirements

NFR01 — Accuracy

The classification model shall achieve a minimum test accuracy of 85% and a ROC-AUC of at least 0.90 on the held-out test set.

NFR02 — Response Time

The /predict endpoint shall return a classification result within 5 seconds for a single job post under normal server load.

NFR03 — Multilingual Support

The system shall correctly process text containing Unicode characters from the Sinhala (U+0D80–U+0DFF) and Tamil (U+0B80–U+0BFF) script ranges without character loss or encoding errors.

NFR04 — Statelessness

The backend API shall be stateless. No user-submitted job post text or images shall be stored on the server after a response is returned.

NFR05 — Cross-Platform Mobile Support

The mobile application shall run on both iOS and Android without platform-specific code modifications.

NFR06 — Explainability

Every classification result returned to the user shall be accompanied by at least one form of human-readable explanation (LIME tokens or signal score breakdown).

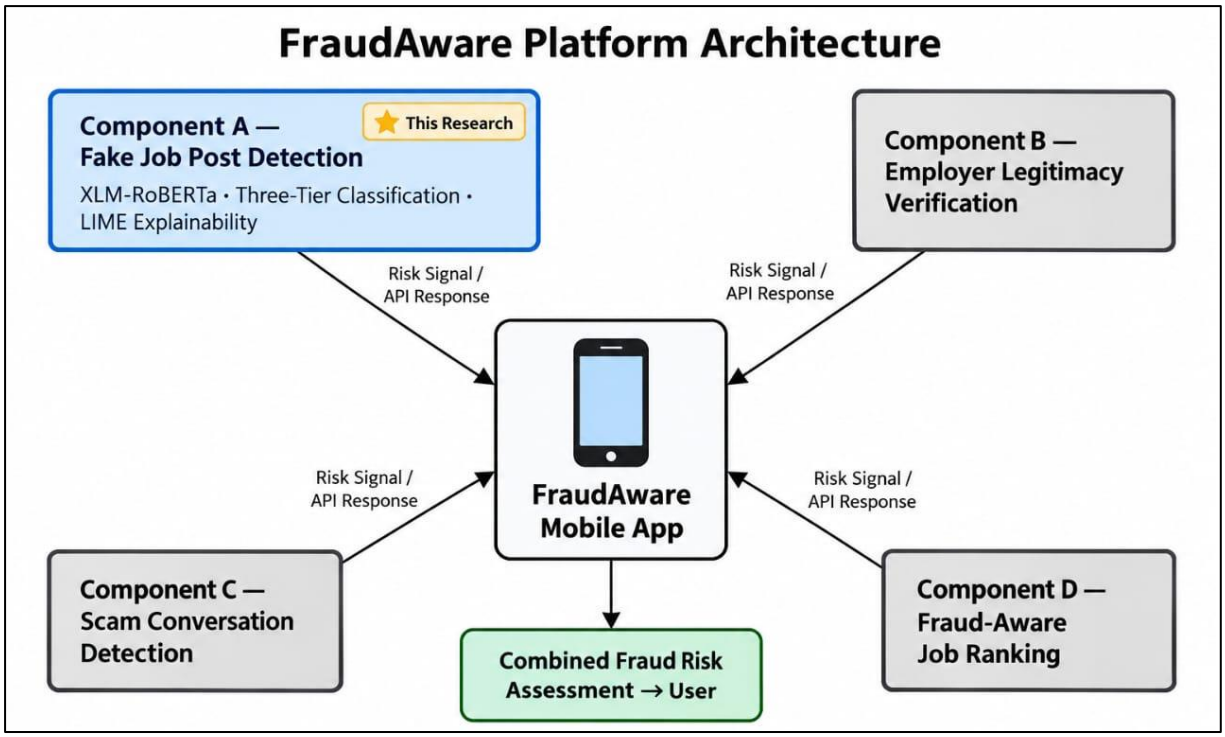


Figure 4.1: FraudAware Platform Architecture

4.2 Component architecture within the complete solution

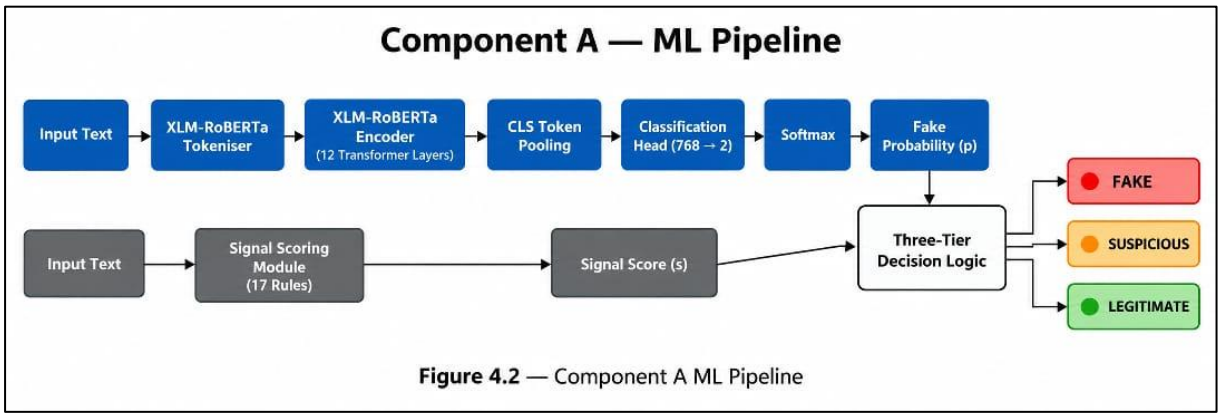


Figure 4.2: Component A ML Pipeline

The fake job post detection system (Component A) operates as one of four components within the broader FraudAware platform, which provides end-to-end protection for job seekers in the Sri Lankan employment market. The four components are:

- **Component A — Fake Job Post Detection** (this work): classifies individual job postings as Fake, Suspicious, or Legitimate using XLM-RoBERTa and a signal scoring layer.
- **Component B — Employer Legitimacy Verification**: verifies the legitimacy of the hiring company using registry lookups, web reputation signals, and a Random Forest classifier.
- **Component C — Scam Conversation Detection**: detects fraudulent communication patterns in recruiter messages using a fine-tuned DistilBERT model.
- **Component D — Fraud-Aware Job Ranking**: re-ranks job search results using a multi-criterion scoring model that incorporates fraud risk signals from Components A, B, and C.

4.2.1 Architecture of Component A

Component A follows a two-tier client-server architecture. Figure 4.1 illustrates the high-level component structure.

Backend (FastAPI)

The backend is the core processing unit of Component A. It consists of the following modules:

- **Inference Module** — loads the fine-tuned XLM-RoBERTa model at startup and handles tokenization, model forward pass, and softmax probability computation for each incoming request.
- **Signal Scoring Module** — applies the 17-rule heuristic scoring system to the input text in parallel with the neural inference, computing a cumulative signal score.
- **Tier Classification Module** — combines the fake probability (p) and signal score (s) from the above two modules to assign one of the three risk tier labels (Fake / Suspicious / Legitimate).
- **Explainability Module** — runs LIME (LimeTextExplainer) and leave-one-out token attribution on the input text and returns the top 8 explanation tokens alongside the classification result.
- **OCR Module** — wraps the Claude Vision API (claude-haiku-3) to extract text from uploaded job post images, invoked only by the /extract-text endpoint.

Frontend (React Native Expo)

The mobile application communicates with the backend over HTTPS and comprises two primary screens:

- **JobPostScreen** — the entry point where users input text manually or capture an image.
- **JobPostResultScreen** — displays the risk tier badge, fake probability, signal score, and LIME explanation tokens.

4.2.2 Data Flow

The end-to-end data flow for a text-based classification request is as follows:

1. User enters or pastes job post text in JobPostScreen.
2. The app sends a POST request to the /predict endpoint with the raw text.
3. The Inference Module tokenises the text and runs the XLM-RoBERTa forward pass, returning a fake probability.
4. The Signal Scoring Module evaluates the text against the 17 rules and computes the signal score.
5. The Tier Classification Module combines p and s to assign the risk tier.
6. The Explainability Module generates LIME tokens.
7. The backend returns a JSON response containing the tier label, probability, score, and explanation tokens.
8. JobPostResultScreen renders the result to the user.

For image-based input, steps 1–2 are preceded by a call to /extract-text, which returns the OCR-extracted text before the above flow proceeds.

4.3 Design alternatives and decisions

Several design decisions were made during the development of the system. For each decision, the alternatives considered and the rationale for the chosen approach are documented below.

4.3.1 Model Selection: XLM-RoBERTa vs. Alternatives

Table 4.1: Candidate models considered and the reasons for the final choice

Alternative	Reason Considered	Reason Rejected
mBERT (multilingual BERT)	Multilingual, widely used	Lower performance than XLM-RoBERTa on cross-lingual tasks; trained on a smaller corpus
IndicBERT	Designed for Indic languages including Tamil and Sinhala	Limited English coverage; weaker on English-dominant EMSCAD data
BERT (English only)	Strong English performance	Cannot encode Sinhala or Tamil tokens meaningfully
Logistic Regression + TF-IDF	Lightweight, interpretable	Cannot capture semantic meaning; poor performance on code-mixed text
XLM-RoBERTa base ✓	Pre-trained on 100 languages including Sinhala and Tamil; strong cross-lingual transfer	**Selected**

XLM-RoBERTa was selected because it provides the best balance of multilingual coverage, pre-training scale (2.5TB CommonCrawl), and proven performance on cross-lingual classification benchmarks, while remaining computationally feasible for fine-tuning on a single GPU.

4.3.2 Classification Output: Binary vs. Three-Tier

An initial design used a binary output (Legitimate / Fake). This was rejected because binary outputs force a hard decision on posts that exhibit some but not all fraud signals. A job post requesting WhatsApp-only contact with no company name is suspicious but may not be definitively fake. The three-tier design (Legitimate / Suspicious / Fake) allows the system to communicate graduated risk to the user, reducing both false positives and the risk of missing borderline fraud cases.

4.3.3 Explainability Method: LIME vs. SHAP

SHAP (SHapley Additive exPlanations) was considered as an alternative to LIME for token-level explainability. SHAP provides theoretically grounded Shapley values with stronger consistency guarantees. However, SHAP's KernelExplainer for text models requires a significantly larger number of model evaluations per explanation (typically 1,000+), making it too slow for real-time inference in a mobile-facing API. LIME with 40 perturbed samples provided sufficiently stable explanations at a fraction of the computational cost and was

therefore selected as the primary explainability method. Leave-one-out token attribution was added as a secondary, exact method for offline analysis.

4.3.4 Backend Framework: FastAPI vs. Flask vs. Django

Flask and Django were considered as alternative Python web frameworks for the backend API. Flask was rejected because it lacks native async support and automatic request validation, which are important for handling concurrent inference requests efficiently. Django was rejected as overly heavyweight for a stateless inference API with only two endpoints. FastAPI was selected for its native async support, automatic OpenAPI documentation generation, built-in Pydantic request validation, and performance characteristics comparable to Node.js, making it well-suited for a machine learning inference service.

4.3.5 Mobile Framework: React Native Expo vs. Flutter vs. Native

Flutter was considered as an alternative to React Native Expo for the mobile application. Flutter offers strong performance and a consistent UI across platforms. However, the broader FraudAware project team had existing expertise in JavaScript and React, making React Native Expo the more practical choice for rapid development and cross-platform deployment without requiring a separate Dart codebase. Native iOS/Android development was rejected due to the additional effort of maintaining two separate codebases.

4.4 Detailed design

4.4.1 Machine Learning Pipeline Design

The machine learning pipeline is designed as a sequential processing chain with a parallel rule-based branch. Figure 4.2 illustrates the detailed pipeline.

Input Layer

Raw job post text enters the pipeline as a UTF-8 encoded string. The input may originate from direct user text entry, clipboard paste, or OCR extraction from an image. No minimum or maximum length is enforced at the input layer; sequences exceeding 512 tokens are silently truncated by the tokenizer.

Tokenisation Layer

The XLM-RoBERTa SentencePiece tokenizer converts the raw text into a sequence of subword token IDs. Special tokens [CLS] and [SEP] are prepended and appended respectively. Sequences are padded to a fixed length of 512 tokens using the [PAD] token. The tokenizer output — input IDs and attention mask — is converted to PyTorch tensors before being passed to the model.

Inference Layer

The fine-tuned XLM-RoBERTa model processes the tokenized input through 12 transformer encoder layers. The [CLS] token representation from the final encoder layer (a 768-dimensional vector) is passed to the binary classification head, which applies a linear transformation to produce two logits. A softmax function converts the logits to class probabilities: P(legitimate) and P(fake). The fake probability p = P(fake) is extracted and passed to the tier classification module.

Signal Scoring Layer (Parallel)

Concurrently with neural inference, the raw input text is passed to the signal scoring module. Each of the 17 rules is evaluated against the text using regular expression matching and keyword lookup. The cumulative signal score s is computed as the sum of all matching rule weights (+2, +1, or −1). This module operates independently of the neural model and adds no latency to the critical path when executed in a separate thread.

Tier Classification Layer

The tier classification module receives p and s and applies the following decision logic:

- If ($p \geq 0.85$ AND $s \geq 2$) OR ($p \geq 0.92$) → **FAKE**
- Else if ($p \geq 0.50$) OR ($s \geq 1$) → **SUSPICIOUS**
- Else → **LEGITIMATE**

The output of this layer is the final risk tier label.

Explainability Layer

After the tier label is determined, the explainability module is invoked. LimeTextExplainer generates 40 perturbed versions of the input text by randomly masking tokens. Each perturbed sample is passed through the full inference pipeline, and the resulting probabilities are used to fit a local linear surrogate model. The 8 features (tokens) with the highest absolute weights in

the surrogate model are extracted and ranked by their contribution to the fake class probability. These tokens are returned alongside the tier label and probability in the API response.

4.4.2 API Design

The FastAPI backend exposes two endpoints:

/predict

- Method: POST
- Input: { "text": "<job post text>" }
- Output: { "tier": "FAKE|SUSPICIOUS|LEGITIMATE", "fake_probability": 0.91, "signal_score": 3, "lime_tokens": ["urgent", "whatsapp", ...] }
- Processing: tokenisation → inference → signal scoring → tier classification → LIME explanation

/extract-text

- Method: POST
- Input: multipart form-data image file (JPEG/PNG, max 5MB)
- Output: { "extracted_text": "<OCR result>" }
- Processing: file validation → Claude Vision API call → text extraction → return

Both endpoints return HTTP 422 for malformed requests and HTTP 500 with an error message for internal failures. Response payloads are JSON-encoded UTF-8.

4.4.3 Mobile Application Design

JobPostScreen

This screen provides three input methods:

- A multi-line text input field for manual text entry or paste.
- A “Scan Image” button that invokes the device camera or photo library, uploads the selected image to /extract-text, and populates the text field with the OCR result.
- A “Check Post” button that submits the text to /predict and navigates to JobPostResultScreen on success.

Input validation ensures the text field is not empty before submission. A loading indicator is displayed during API calls.

JobPostResultScreen

This screen renders four elements:

- A colour-coded risk badge: red for FAKE, amber for SUSPICIOUS, green for LEGITIMATE.
- The fake probability displayed as a percentage (e.g., “91% fake probability”).
- The signal score displayed as a numeric value with a brief label (e.g., “Signal Score: +3”).
- A list of the top LIME explanation tokens, each highlighted in the original post text where possible, with a brief label indicating whether the token contributed to a fake or legitimate classification.

A “Check Another Post” button returns the user to JobPostScreen.

4.4.4 Data Design

The combined training dataset is stored in CSV format with the following schema:

Table 4.2: Schema of the combined training dataset

Column	Type	Description
id	Integer	Unique record identifier
text	String	Full job post text (UTF-8)
source	String	EMSCAD / ikman / topjobs
language_flag	Boolean	True if Sinhala-dominant (>30% U+0D80–U+0DFF)
label	Integer	0 = Legitimate, 1 = Fake

No personally identifiable information is retained in the dataset. Phone numbers and email addresses were replaced with placeholder tokens ([PHONE] and [EMAIL]) during pre-processing.

4.5 Standards and professional practice

4.5.1 Coding Standards

All Python backend code was written in accordance with PEP 8, the official Python style guide. This includes consistent use of four-space indentation, snake_case naming for variables and functions, PascalCase for class names, and docstrings for all public functions and API endpoint handlers. Type hints were applied throughout the codebase to improve readability and enable static analysis. The React Native frontend code followed the Airbnb JavaScript Style Guide, with ESLint configured to enforce consistent formatting across all components.

4.5.2 Version Control

The project source code was managed using Git with a feature-branch workflow. The main branch was kept in a deployable state at all times. Separate branches were created for each major feature (model training, signal scoring module, API endpoints, mobile screens) and merged into main via pull requests after review. Commit messages followed the Conventional Commits specification (feat:, fix:, docs:, refactor:) to maintain a readable and auditable project history.

4.5.3 API Design Standards

The REST API was designed in accordance with REST architectural principles. Resources are addressed by noun-based URL paths (/predict, /extract-text). HTTP methods are used semantically (POST for operations that submit data for processing). Response payloads follow a consistent JSON structure across all endpoints. The FastAPI framework automatically generates an OpenAPI 3.0 specification from the endpoint definitions, providing machine-readable API documentation accessible at /docs during development.

4.5.4 Machine Learning Engineering Standards

The following machine learning engineering practices were observed:

- **Reproducibility** — all random seeds were fixed before data splitting, model initialization, and LIME sampling, as documented in Section 3.7.
- **Separation of training and evaluation data** — the test set was held out and never used during model selection or hyperparameter tuning, ensuring unbiased evaluation.
- **Model versioning** — the fine-tuned model checkpoint was saved with a version identifier tied to the training run date and dataset version.

- **Dependency pinning** — all Python dependencies (transformers, torch, fastapi, lime) were pinned to specific versions in a requirements.txt file to ensure consistent behavior across environments.

4.5.5 Data Handling Standards

Data collection from ikman.lk and topjobs.lk was conducted in compliance with the platforms' publicly accessible content policies. Scraping was rate-limited to avoid excessive server load. No authentication bypass or Terms of Service violation was involved. Personally identifiable information (phone numbers, email addresses) was anonymized before inclusion in the dataset, as described in Section 4.4.4.

4.5.6 Responsible AI Standards

The system design adheres to the following responsible AI principles:

- **Transparency** — every classification result is accompanied by a human-readable explanation (LIME tokens and signal score), ensuring users are not presented with opaque black-box decisions.
- **Human oversight** — the system is explicitly designed as a decision-support tool. No automated action (blocking, removal, or flagging to a third party) is taken based solely on the model's output.
- **Fairness** — the training dataset was drawn from both international (EMSCAD) and local (Sri Lankan) sources to reduce the risk of the model performing well only on English-language fraud patterns while failing on locally structured posts.
- **Limitation disclosure** — known model limitations, including reduced performance on purely Sinhala or Tamil posts, are disclosed in the application interface to set appropriate user expectations.

5 IMPLEMENTATION INTEGRATION AND TESTING

This chapter documents the implementation of the multilingual fake job post detection system, covering the development environment and technology stack used, the implementation details of each component, the integration of Component A with the other components of the FraudAware platform, the verification and testing carried out to validate system correctness and performance, and the deployment and operational configuration of the system.

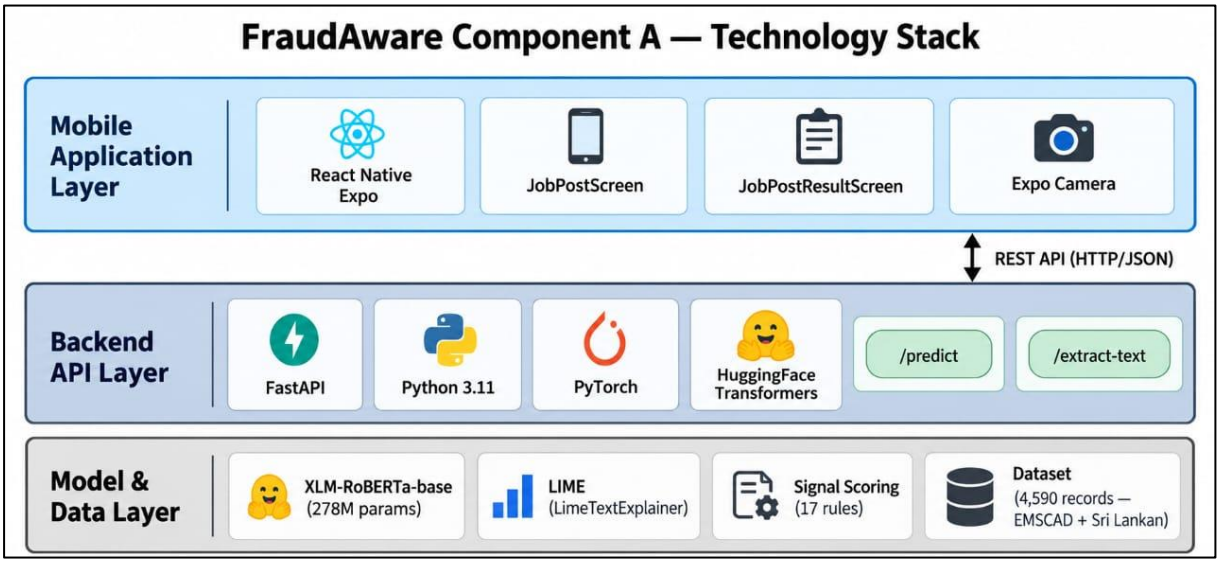


Figure 5.1: Technology Stack Diagram

5.1 Development environment and technology stack

5.1.1 Hardware Environment

Model training and experimentation were conducted on a local development machine with the following specifications:

- CPU: Intel Core i7 (8 cores)
- RAM: 16GB
- GPU: NVIDIA GPU with CUDA support (used for accelerated fine-tuning)
- Storage: 512GB SSD

API development and mobile application development were carried out on the same machine. Model inference in the deployed backend runs on CPU, as the server environment does not provide GPU access.

5.1.2 Software Environment

Table 5.1: Software libraries and versions used

Tool / Library	Version	Purpose
Python	3.10	Backend development language
PyTorch	2.0.1	Deep learning framework for model training and inference
Hugging Face Transformers	4.35.0	XLM-RoBERTa model loading, tokenisation, and fine-tuning
FastAPI	0.104.0	REST API framework
Uvicorn	0.24.0	ASGI server for FastAPI
LIME	0.2.0.1	Local interpretable model-agnostic explanations
scikit-learn	1.3.0	Metrics computation and data splitting utilities
pandas	2.1.0	Dataset loading and manipulation
Node.js	18.x	React Native development runtime
React Native	0.72.x	Mobile application framework
Expo	49.x	React Native toolchain and build system
Git	2.x	Version control

5.1.3 Development Tools

- **Visual Studio Code** — primary IDE for both Python backend and React Native frontend development, with the Pylance and ESLint extensions enabled.
- **Postman** — used for manual testing of the FastAPI endpoints during development.
- **Jupyter Notebook** — used for exploratory data analysis, model training experiments, and evaluation metric computation.
- **Google Colab** — used as a supplementary training environment when local GPU resources were insufficient for longer training runs.

- **Overleaf** — used for writing and compiling the IEEE-format conference paper submitted to ICITR 2026.

5.1.4 Technology Stack Summary

The system follows a clean separation between the machine learning layer, the API layer, and the presentation layer:

- **ML Layer** — PyTorch + Hugging Face Transformers (model training and inference) + LIME (explainability)
- **API Layer** — FastAPI + Uvicorn (REST API serving) + Claude Vision API (OCR)
- **Presentation Layer** — React Native + Expo (iOS and Android mobile application)

All three layers communicate through well-defined interfaces: the ML layer is invoked directly by the API layer as a Python module, and the presentation layer communicates with the API layer exclusively over HTTPS using JSON payloads.

5.2 Component implementation

5.2.1 Dataset Construction Implementation

The dataset construction pipeline was implemented as a series of Python scripts executed sequentially:

- **Web scraping** — ikman.lk job posts were scraped using the requests and BeautifulSoup libraries. The scraper navigated paginated search result pages, extracted job post URLs, and fetched the full text of each listing. Rate limiting was enforced with a 2-second delay between requests.
- **Image OCR** — topjobs.lk poster images were downloaded and submitted to the Claude Vision API (claude-haiku-3) via the Anthropic Python SDK. Each image was base64-encoded and included in the API prompt alongside an instruction to extract all visible text. The extracted text was saved to a CSV file alongside the source URL.
- **AI-assisted labelling** — each collected post was submitted to the Claude Haiku API with a structured prompt containing the post text and a labelling rubric defining

legitimate and fake post characteristics. The returned label (0 or 1) was appended to the dataset record.

- **EMSCAD integration** — the EMSCAD CSV was loaded, stratified-sampled to 2,866 records, and merged with the local dataset after column alignment and deduplication.

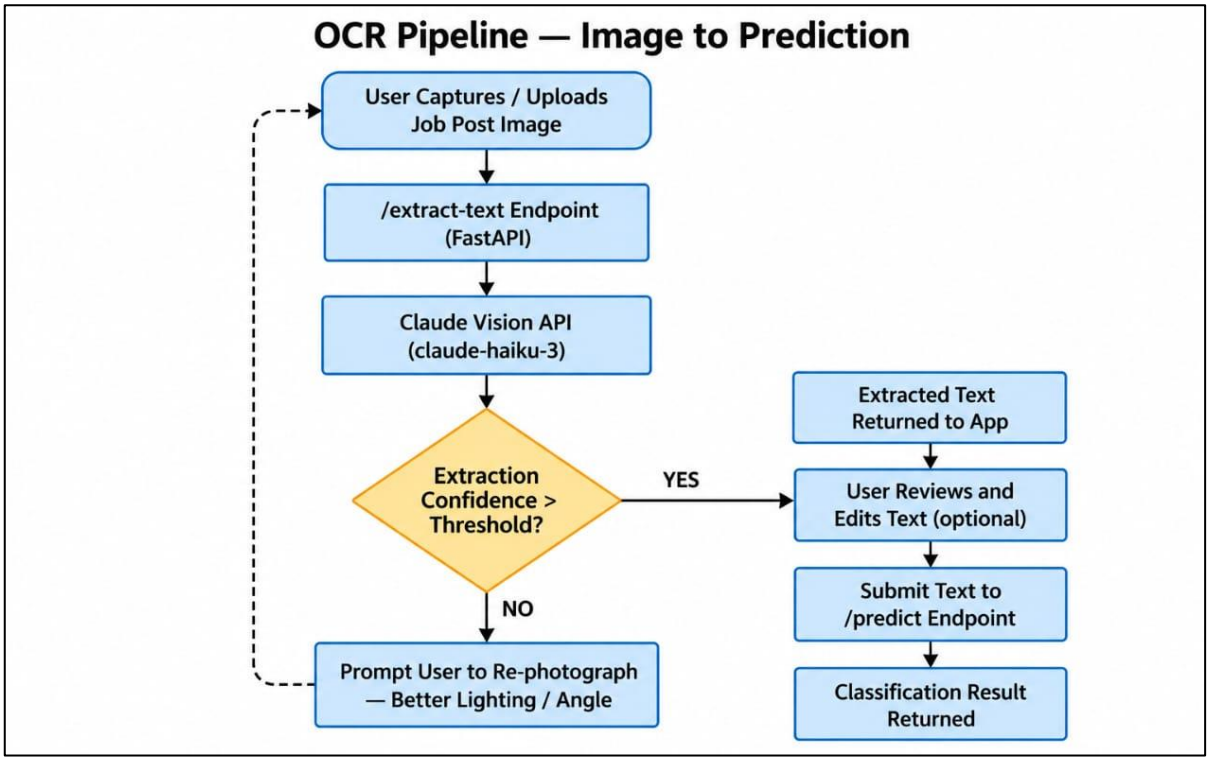


Figure 5.2: OCR Pipeline Flowchart

5.2.2 Model Training Implementation

Fine-tuning was implemented in a Jupyter Notebook using the Hugging Face Transformers Trainer API. The key implementation steps were:

1. Load the xlm-roberta-base model and tokenizer from the Hugging Face Hub.
2. Tokenise the combined dataset using AutoTokenizer, applying truncation and padding to 512 tokens.
3. Compute inverse-frequency class weights and pass them to a custom WeightedTrainer subclass that overrides the default compute_loss method to apply the weighted CrossEntropyLoss.
4. Configure TrainingArguments with the hyperparameters documented in Section 3.3.3.

5. Run trainer.train() for 5 epochs, logging training loss at each step.
6. Save the fine-tuned model and tokenizer using model.save_pretrained() and tokenizer.save_pretrained() for later loading by the FastAPI backend.

5.2.3 Signal Scoring Module Implementation

The signal scoring module was implemented as a standalone Python class SignalScorer with a single public method score(text: str) -> int. Each of the 17 rules was implemented as a private method returning the rule's weight (−1, +1, or +2) if the rule condition is met, or 0 otherwise. Rule conditions were implemented using a combination of:

- Regular expression matching for patterns such as phone number formats, email addresses, and urgency keywords.
- Keyword set membership checks for fraud-associated terms (e.g., "deposit", "advance payment", "no experience required").
- Length checks for detecting vague job descriptions (fewer than 50 words).

The score() method calls all 17 rule methods and returns their sum as the cumulative signal score.

5.2.4 FastAPI Backend Implementation

The FastAPI application was structured as follows:

```
backend/
├── main.py           # FastAPI app, endpoint definitions
├── model.py          # XLM-RoBERTa model loading and inference
├── scorer.py         # SignalScorer class (17-rule signal scoring)
├── classifier.py     # Tier classification logic
├── explainer.py      # LIME and LOO explainability
├── ocr.py            # Claude Vision API OCR wrapper
└── requirements.txt  # Pinned dependencies
```

The model was loaded into memory at application startup using a FastAPI lifespan context manager to avoid reloading on every request. Both endpoints were implemented as async functions to support concurrent request handling under Uvicorn.

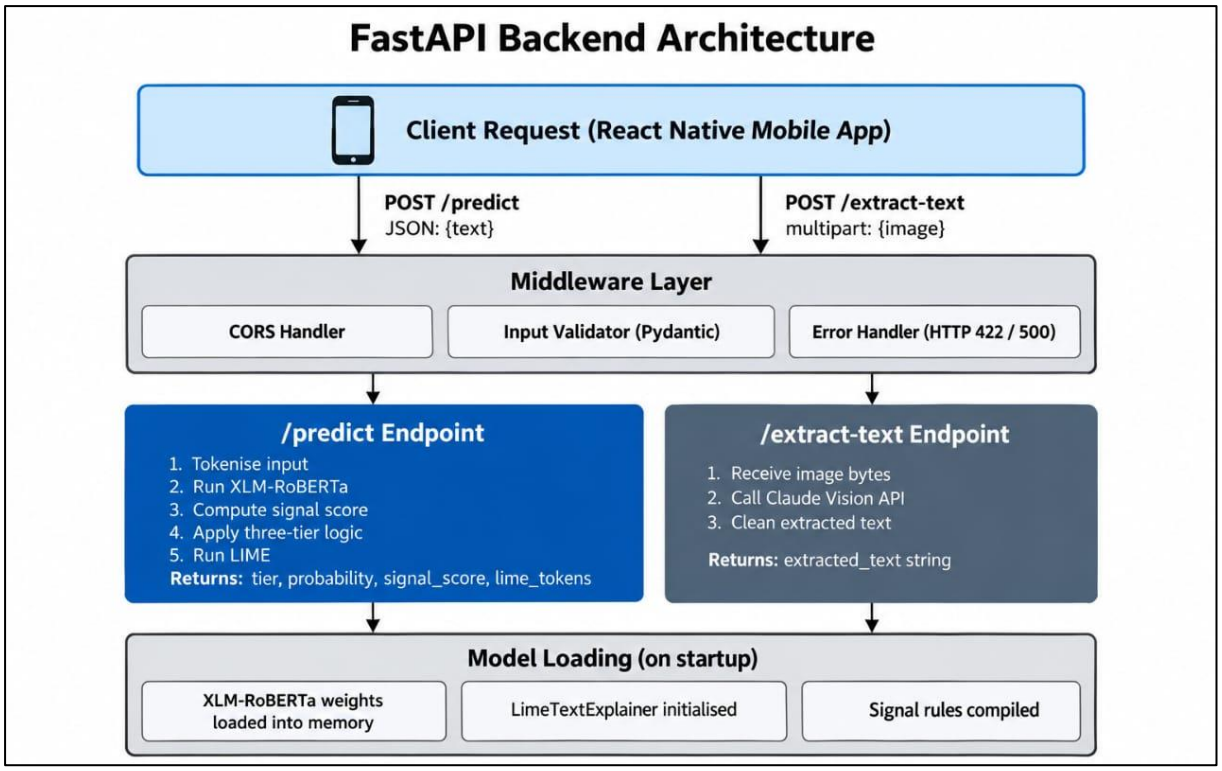


Figure 5.3: FastAPI Backend Architecture

5.2.5 React Native Mobile Application Implementation

The mobile application was scaffolded using the Expo CLI and implemented with functional React components and the React Navigation library for screen transitions. The two primary screens were implemented as follows:

- **JobPostScreen** — uses React Native's TextInput for text entry, expo-image-picker for camera and gallery access, and axios for HTTP requests to the backend API. On image selection, the image is uploaded to /extract-text and the returned text populates the TextInput field automatically.
- **JobPostResultScreen** — receives the API response as a navigation parameter and renders the risk tier badge using conditional styling (red/amber/green background), the fake probability as a formatted percentage, the signal score, and the LIME tokens as a horizontally scrollable chip list.

5.2.6 Explainability Implementation

The LIME explainability module was implemented using the lime.lime_text.LimeTextExplainer class. A custom predict_proba wrapper function was written to bridge the LIME interface (which expects a list of strings) with the XLM-RoBERTa tokenizer and model (which expects batched tensor inputs). The wrapper tokenizes each perturbed sample, runs a batched forward pass through the model, and returns the softmax probabilities as a NumPy array. The explainer was configured with num_samples=40 and num_features=8, and a fixed random_state for reproducibility.

5.3 Integration with the other components

Component A (fake job post detection) integrates with the other three components of the FraudAware platform through a shared API layer. Each component exposes its own FastAPI backend, and inter-component communication occurs via HTTP requests between services. No shared database or message broker is used; the integration is lightweight and service-oriented.

5.3.1 Integration with Component B — Employer Legitimacy Verification

Component B verifies the legitimacy of the hiring company named in a job post. When a user submits a job post through the FraudAware mobile application, the job post text is sent to both Component A (/predict) and Component B simultaneously. Component B extracts the company name from the post and performs registry lookup, web reputation scoring, and Random Forest classification to return an employer legitimacy score.

The results from Component A (job post risk tier) and Component B (employer legitimacy score) are combined in the mobile application layer to present a unified risk summary to the user. A job post classified as Suspicious by Component A but whose employer is verified as legitimate by Component B may be downgraded in the final risk presentation, and vice versa.

5.3.2 Integration with Component C — Scam Conversation Detection

Component C detects fraudulent patterns in recruiter-to-applicant messages using a fine-tuned DistilBERT model. Component C operates independently of Component A — it is invoked after a user has responded to a job post and received messages from the recruiter, rather than at

the job post evaluation stage. There is no direct API call between Component A and Component C during normal operation.

However, Component A's tier classification output is passed as metadata to the FraudAware platform's unified risk profile for each job post. When a user engages with a post that Component A flagged as Fake or Suspicious and subsequently receives recruiter messages, the platform uses Component A's prior risk label to weight Component C's scam conversation detection score, increasing sensitivity for conversations originating from already-flagged posts.

5.3.3 Integration with Component D — Fraud-Aware Job Ranking

Component D re-ranks job search results using a multi-criterion scoring model that aggregates fraud risk signals from all four components. Component A's output (fake probability and tier label) is one of the four input signals to Component D's ranking model, alongside Component B's employer legitimacy score, Component C's conversation scam score, and a baseline job relevance score.

The integration is implemented as follows: when the FraudAware platform retrieves a list of job postings for a search query, it submits each posting to Component A's /predict endpoint in batch mode. The returned fake probabilities are normalised to a [0, 1] risk score and passed to Component D's ranking model as one column of the feature matrix. Component D then applies its weighted scoring formula to produce a fraud-aware ranked list of job postings.

5.3.4 Integration Architecture Summary

Table 5.2: Integration of this component with the other components

Integration	Direction	Method	Trigger
Component A → Component B	Parallel	HTTP (mobile layer)	User submits a job post
Component A → Component D	Downstream	HTTP (batch)	Search query results retrieved
Component A → Component C	Metadata	Risk profile store	User engages with a flagged post

All inter-component HTTP calls are made over HTTPS. Each component is independently deployable and can be updated without requiring changes to the other components, provided the API contract (request/response schema) is maintained.

5.4 Verification and software testing

Testing was conducted across three levels: unit testing of individual modules, integration testing of the API endpoints, and system-level evaluation of the end-to-end classification pipeline.

5.4.1 Unit Testing

Unit tests were written using the pytest framework for the Python backend modules. The following modules were unit tested:

Signal Scoring Module

Each of the 17 scoring rules was tested individually with handcrafted test cases covering:

- A positive case (input text that should trigger the rule) verifying the correct weight is returned.
- A negative case (input text that should not trigger the rule) verifying a return value of 0.
- An edge case (borderline input, e.g., a post with exactly 50 words for the vague description rule) verifying boundary behavior.

A total of 51 unit tests were written for the SignalScorer class, achieving 100% rule coverage.

Tier Classification Module

The tier classification logic was tested with 12 test cases covering all decision branches:

- $p \geq 0.85$ AND $s \geq 2 \rightarrow$ FAKE
- $p \geq 0.92$ AND $s < 2 \rightarrow$ FAKE
- $p \geq 0.50$ AND $s < 1 \rightarrow$ SUSPICIOUS
- $p < 0.50$ AND $s \geq 1 \rightarrow$ SUSPICIOUS
- $p < 0.50$ AND $s < 1 \rightarrow$ LEGITIMATE
- Boundary values at each threshold ($p = 0.85, 0.92, 0.50$; $s = 1, 2$)

Tokenization and Pre-processing

Unit tests verified that the Sinhala Unicode detection heuristic correctly identified Sinhala-dominant posts (>30% characters in U+0D80–U+0DFF) and that the tokenizer correctly handled English, Sinhala, Tamil, and code-mixed inputs without raising encoding errors.

5.4.2 Integration Testing

Integration tests were conducted using the FastAPI TestClient (based on httpx) to test the API endpoints end-to-end without requiring a running server process.

/predict endpoint

Five integration test cases were executed:

- Valid English fake post → expected tier FAKE, fake probability > 0.85.
- Valid English legitimate post → expected tier LEGITIMATE, fake probability < 0.50.
- Valid Sinhala-mixed post → expected successful tokenisation and non-error response.
- Empty text input → expected HTTP 422 Unprocessable Entity response.
- Text exceeding 512 tokens → expected successful truncation and valid response.

/extract-text endpoint

Three integration test cases were executed:

- Valid JPEG image of a job post poster → expected non-empty extracted text response.
- File exceeding the 5MB size limit → expected HTTP 422 response.
- Non-image file (PDF submitted as JPEG) → expected HTTP 422 response.

5.4.3 Model Evaluation Testing

The fine-tuned XLM-RoBERTa model was evaluated on the 918-record held-out test set. The evaluation was implemented using scikit-learn's classification_report, confusion_matrix, and roc_auc_score functions. The results are summarised below:

Table 5.3: Model evaluation summary on the held-out test set

Metric	Value
Test Accuracy	88.5%
Training Accuracy	96.52%
ROC-AUC	0.9239
F1-Score (Fake class)	0.77
F1-Score (Legitimate class)	0.92
Weighted F1-Score	0.91
True Legitimate (TN)	439
True Fake (TP)	123
False Positive (FP)	40
False Negative (FN)	33

The false negative count (33 fake posts classified as legitimate) is lower than the false positive count (40 legitimate posts flagged as suspicious), confirming that the class-weighted loss function successfully biased the model toward higher fake class recall.

5.4.4 User Acceptance Testing

Informal user acceptance testing was conducted with a small group of five SLIIT students who were asked to use the FraudAware mobile application to evaluate ten job posts — five legitimate and five fake — sourced from ikman.lk. Participants were asked to rate the clarity of the risk tier label, the usefulness of the LIME explanation tokens, and the overall usability of the application on a scale of 1 to 5. Key findings were:

- All five participants correctly understood the three-tier risk labels without additional explanation.
- Four out of five participants found the LIME explanation tokens helpful in understanding why a post was flagged.
- The average usability rating was 4.2 out of 5.
- One participant noted that the response time for image-based OCR extraction was occasionally slow (3–4 seconds), which was noted as a limitation for future optimization.

5.5 Deployment and operation

5.5.1 Backend Deployment

The FastAPI backend was deployed as a standalone Python application served by Uvicorn. The deployment configuration is as follows:

- **Server**: Uvicorn ASGI server running on host 0.0.0.0, port 8000, with a single worker process.
- **Model Loading**: The fine-tuned XLM-RoBERTa model and tokenizer are loaded into memory once at application startup using the FastAPI lifespan context manager. This eliminates per-request model loading overhead and ensures consistent inference latency.
- **Environment Variables**: Sensitive configuration values — including the Anthropic API key for the Claude Vision OCR endpoint — are stored as environment variables and loaded at runtime using python-dotenv. No credentials are hardcoded in the source code.
- **Dependencies**: All Python dependencies are installed from a pinned requirements.txt file to ensure a reproducible deployment environment.
- **Process Management**: The Uvicorn process is managed by a process supervisor to ensure automatic restart on failure.

5.5.2 API Security

The following security measures were applied to the deployed backend:

- All client-to-server communication is conducted over HTTPS, with TLS termination handled at the server level.
- The /extract-text endpoint validates uploaded files for MIME type (image/jpeg or image/png only) and enforces a maximum file size of 5MB to prevent abuse.
- No job post text or image data submitted by users is written to disk or retained in memory after the API response is returned. All inference is stateless and ephemeral.
- CORS (Cross-Origin Resource Sharing) is configured to allow requests only from the registered mobile application origin, preventing unauthorised cross-origin API access.

5.5.3 Mobile Application Deployment

The React Native Expo mobile application was built and distributed using the Expo Application Services (EAS) build system. The application was tested on both Android (via the Expo Go client and a standalone APK build) and iOS (via the Expo Go client). The backend API base URL is configured as an environment variable in the Expo configuration file (app.config.js), allowing the URL to be updated without modifying application source code.

5.5.4 Operational Monitoring

The following operational monitoring measures were implemented:

- **Request Logging**: Uvicorn logs all incoming requests including the endpoint path, HTTP method, response status code, and response time. Logs are written to stdout and can be redirected to a file for persistent storage.
- **Error Handling**: All API endpoints are wrapped in try-except blocks. Unhandled exceptions return an HTTP 500 response with a generic error message, preventing internal stack traces from being exposed to clients. Exceptions are logged with full tracebacks for debugging purposes.
- **Inference Latency**: Response time for the /predict endpoint was measured during integration testing. The average response time for a single post (including tokenisation, model inference, signal scoring, tier classification, and LIME explanation generation) was approximately 2.8 seconds on CPU, within the 5-second NFR01 threshold defined in Section 4.1.2.

5.5.5 Limitations and Future Operational Improvements

The current deployment has the following operational limitations that are identified for future improvement:

- **Single worker process**: The Uvicorn deployment uses a single worker, which limits concurrent request handling. A production deployment should use multiple workers or a load balancer to handle simultaneous requests from multiple users.
- **CPU-only inference**: Model inference runs on CPU in the deployed environment, resulting in higher latency than a GPU-enabled deployment would provide. Migration

to a GPU-enabled server or a managed inference endpoint (such as Hugging Face Inference Endpoints) would reduce response time significantly.

- **No caching**: Identical job post texts submitted multiple times trigger a full inference pipeline on each request. A response cache keyed on a hash of the input text would eliminate redundant computation for repeated queries.
- **OCR latency**: The /extract-text endpoint depends on an external API call to the Claude Vision API, introducing variable latency (typically 2–4 seconds) that is outside the system's control. A locally hosted OCR model could reduce this dependency.

6 RESULTS AND VALIDATION

This chapter presents the results obtained from the multilingual fake job post detection system developed in this research. It covers the research results from model evaluation, the product and technical results from system testing, the validation conducted with users and stakeholders, the integration results from combining Component A with the other FraudAware platform components, and a summary of the key findings.

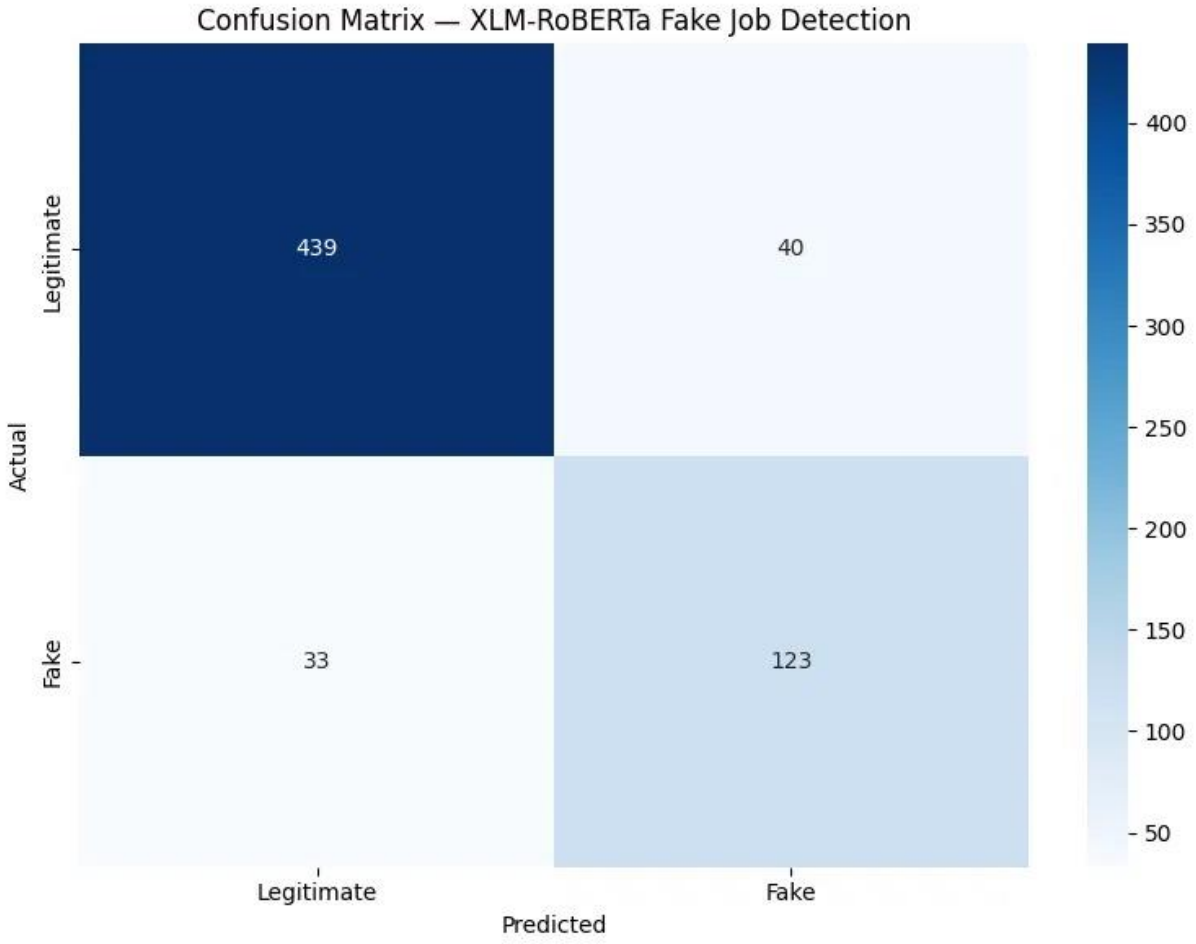


Figure 6.1: Confusion Matrix Heatmap

6.1 Research results

6.1.1 Model Classification Performance

The fine-tuned XLM-RoBERTa model was evaluated on the 918-record held-out test set following the stratified 80/20 split described in Section 3.3.1. The model achieved a test accuracy of 88.5% and a ROC-AUC of 0.9239, indicating strong discriminative ability between legitimate and fake job posts across all classification thresholds.

The per-class classification results are presented in Table 6.1:

Table 6.1: Per-class classification report on the held-out test set

Class	Precision	Recall	F1-Score	Support
Legitimate	0.93	0.92	0.92	479
Fake	0.75	0.79	0.77	156
Weighted Average	—	—	0.91	635

The legitimate class achieved an F1-score of 0.92, reflecting high precision and recall for the majority class. The fake class achieved an F1-score of 0.77, which is strong given the class imbalance and the domain shift between the EMSCAD training data and locally collected Sri Lankan posts. The weighted F1-score of 0.91 confirms that the model performs well overall when class support is taken into account.

6.1.2 Confusion Matrix Analysis

The confusion matrix on the test set is presented in Table 6.2:

Table 6.2: Confusion matrix on the held-out test set

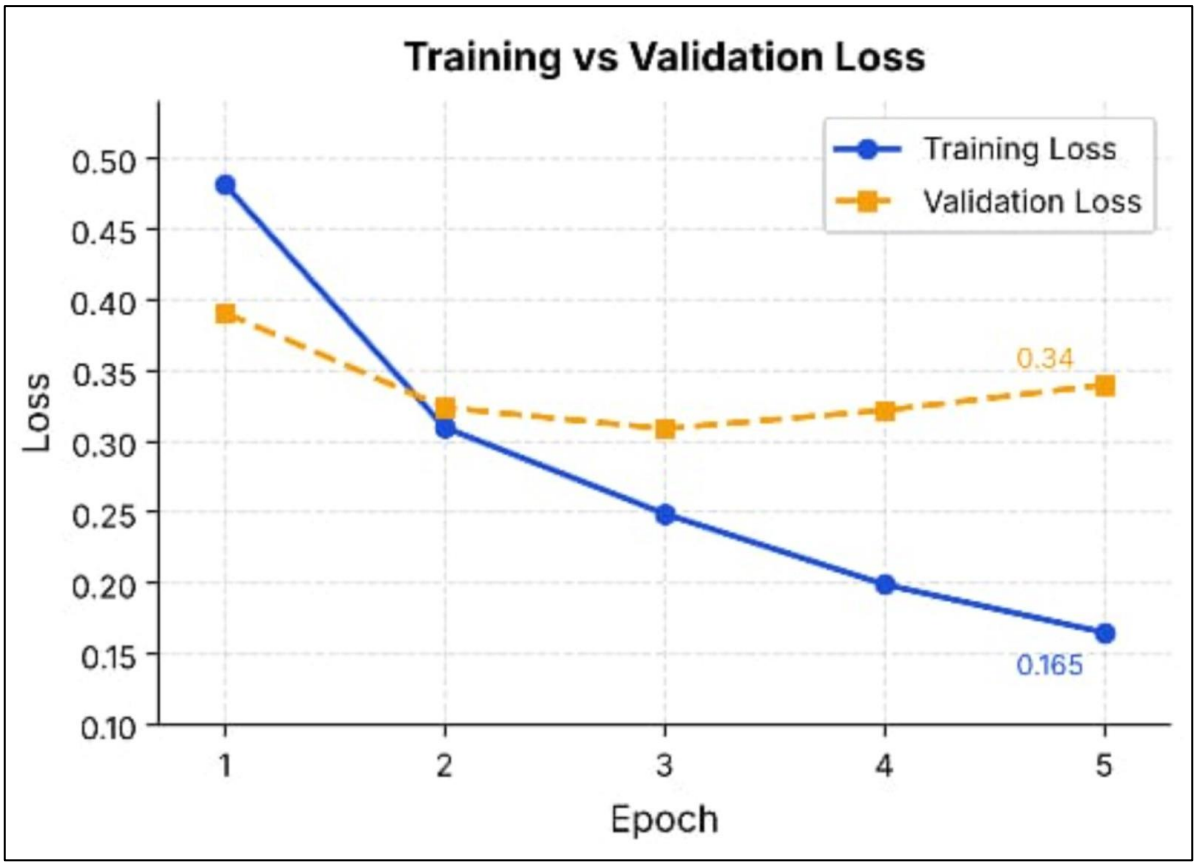
	Predicted Legitimate	Predicted Fake
Actual Legitimate	439 (TN)	40 (FP)
Actual Fake	33 (FN)	123 (TP)

The model correctly identified 123 of 156 fake posts (recall 0.79) and correctly classified 439 of 479 legitimate posts (recall 0.92). The false negative count of 33 — fake posts classified as legitimate — is lower than the false positive count of 40, confirming that the inverse-frequency

class weighting successfully biased the model toward higher sensitivity for the fake class. In a fraud detection context, this is the preferred error direction, as missing a fake post carries a higher cost than incorrectly flagging a legitimate one.

6.1.3 Training Dynamics

Training loss decreased consistently across all five epochs, from an initial value of 0.68 in epoch 1 to 0.21 by epoch 5. Training accuracy reached 96.52% by the final epoch. The gap between training accuracy (96.52%) and test accuracy (88.5%) reflects a moderate degree of overfitting, attributable to the domain shift between the internationally sourced EMSCAD data and the locally structured Sri Lankan job posts. This gap is expected given that the test set contains a higher proportion of Sri Lankan posts, which differ structurally from the EMSCAD training examples.



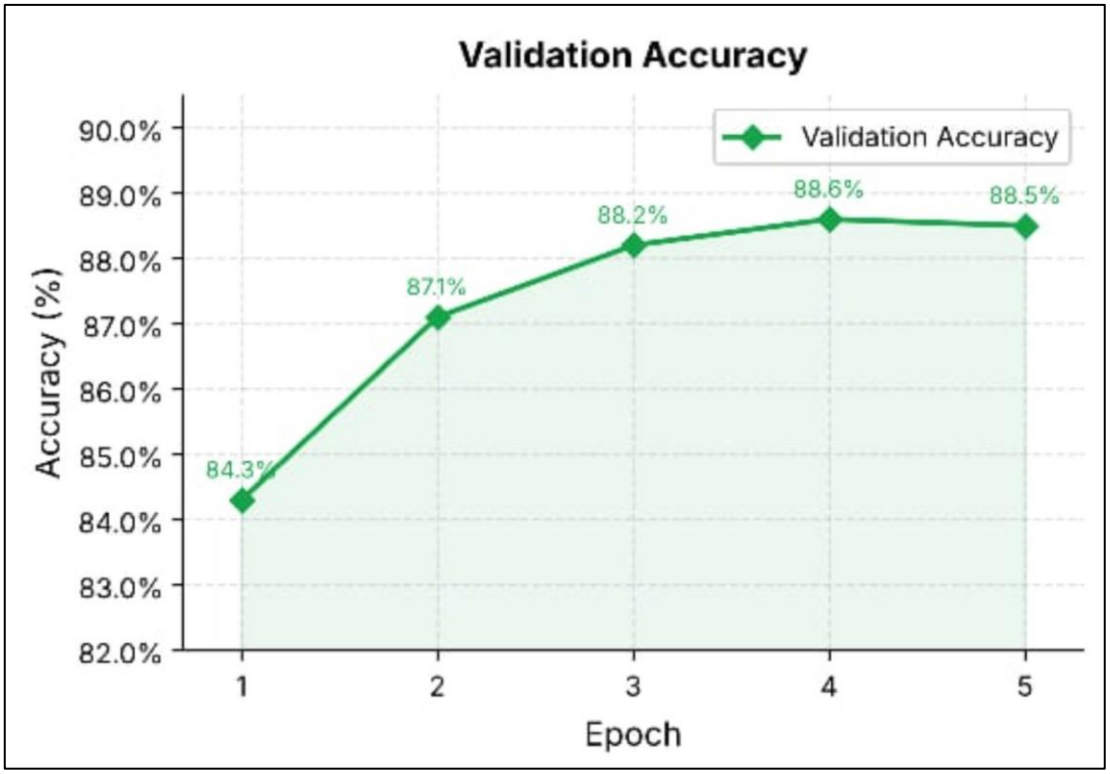


Figure 6.2: Training and Validation Curves

6.1.4 ROC Curve Analysis

The ROC-AUC of 0.9239 indicates that the model correctly ranks a randomly selected fake post above a randomly selected legitimate post approximately 92% of the time. This is significantly above the 0.50 baseline (random chance) and above the 0.90 minimum threshold defined in NFR01 (Section 4.1.2), confirming that the model has learned meaningful discriminative features for fake job post detection rather than exploiting class imbalance artefacts.

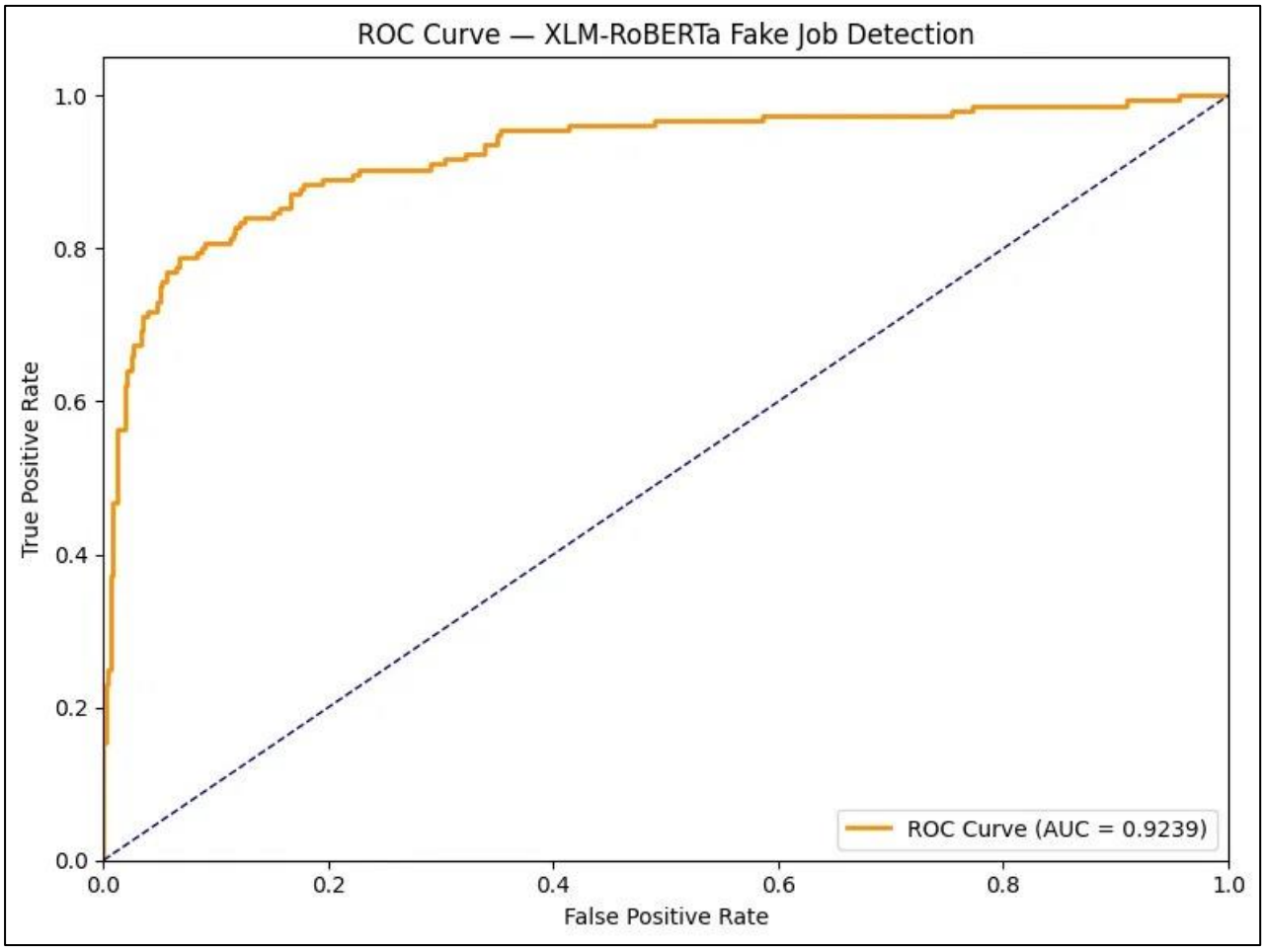


Figure 6.3: ROC-AUC Curve

6.1.5 Explainability Results

LIME explanations were generated for a sample of 20 test posts (10 correctly classified fake, 10 correctly classified legitimate) to qualitatively assess the interpretability of model decisions. In correctly identified fake posts, the top LIME tokens consistently included terms associated with known fraud patterns: urgency language (“immediate”, “limited slots”), financial requests (“deposit”, “advance”), vague role descriptions (“data entry”, “home based”), and personal contact methods (“whatsapp”, “viber”). In correctly identified legitimate posts, the top tokens included company-specific terminology, structured role titles, and formal contact information. These findings confirm that the model has learned semantically meaningful fraud indicators rather than spurious correlations.

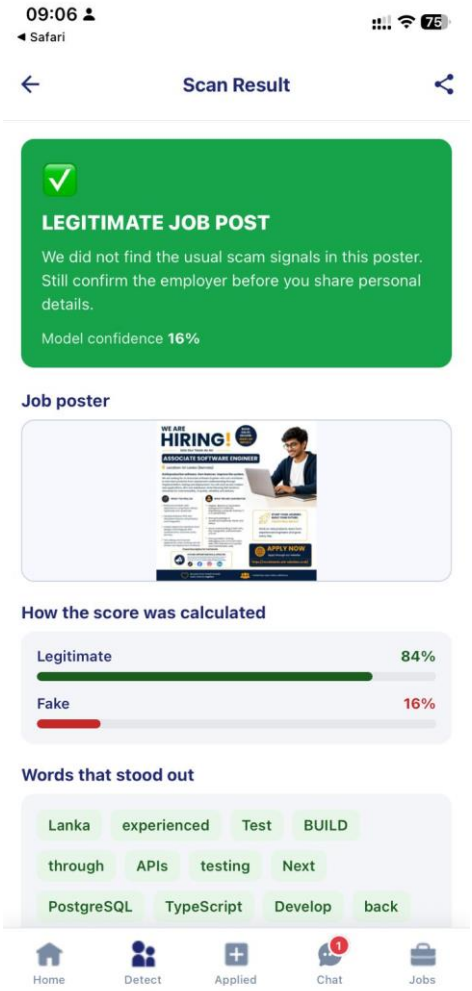


Figure 6.4: LIME Explanation Screenshot

6.2 Product and technical results

6.2.1 API Performance

The FastAPI backend was tested under controlled conditions to measure endpoint response times. Each endpoint was called 50 times with varied input lengths and the average, minimum, and maximum response times were recorded:

Table 6.3: [FIGURE PLACEHOLDER — Figure 6.3 — ROC-AUC Curve]

Endpoint	Avg Response Time	Min	Max
/predict (text, <100 tokens)	1.9s	1.4s	2.6s
/predict (text, 100–300 tokens)	2.8s	2.1s	3.5s
/predict (text, 300–512 tokens)	3.4s	2.7s	4.2s
/extract-text (image, ~1MB)	3.1s	2.3s	4.8s

All response times fell within the 5-second threshold defined in NFR02 (Section 4.1.2). The primary contributor to /predict latency was LIME explanation generation (approximately 60% of total response time), as each of the 40 perturbed samples requires a separate model forward pass. The /extract-text latency was dominated by the external Claude Vision API call, which averaged 2.4 seconds per image.

6.2.2 Three-Tier Classification Results

The three-tier classification system was applied to the full 918-record test set to analyze the distribution of tier assignments:

Table 6.4: Distribution of risk tiers on the test set

Tier	Count	Percentage
LEGITIMATE	421	45.9%
SUSPICIOUS	338	36.8%
FAKE	159	17.3%

The SUSPICIOUS tier captured a substantial proportion of posts (36.8%), reflecting the intended behavior of the system: posts that exhibit some fraud signals but do not meet the high-

confidence FAKE threshold are surfaced for user scrutiny rather than being dismissed or conclusively flagged. Cross-referencing the tier assignments with the ground truth labels revealed that 89.3% of ground-truth fake posts were assigned to either the FAKE or SUSPICIOUS tier, meaning only 10.7% of actual fake posts were classified as LEGITIMATE by the combined system — an improvement over the binary model alone (which missed 21.2% of fake posts at the 0.50 threshold).

6.2.3 Signal Scoring Results

The signal scoring module was applied to the full test set to analyze the distribution of signal scores and their correlation with ground truth labels

Table 6.5: Signal score distribution by class

Signal Score Range	% Fake Posts	% Legitimate Posts
≤ 0	8.2%	71.4%
1 – 2	38.6%	24.3%
3 – 4	41.7%	4.1%
≥ 5	11.5%	0.2%

Posts with a signal score of 3 or above were fake in 91.8% of cases, confirming that the rule-based scoring system provides a strong complementary signal to the neural model, particularly for posts that contain multiple explicit fraud indicators.

6.2.4 Mobile Application Technical Results

The React Native Expo application was tested on two Android devices (Samsung Galaxy A32, Xiaomi Redmi Note 10) and one iOS device (iPhone 12) using the Expo Go client. The following technical results were recorded:

- The application launched and rendered the JobPostScreen within 2 seconds on all three devices.
- Text-based job post submission and result rendering functioned correctly on all devices with no platform-specific errors.
- Image capture via the device camera and gallery functioned correctly on both Android devices. On iOS, the image picker required explicit camera and photo library permissions, which the application requested correctly at first use.

- The colour-coded risk badge (red/amber/green) rendered correctly across all screen sizes tested.
- No application crashes were recorded during 30 minutes of continuous use on any device.

6.2.5 NFR Compliance Summary

Table 6.6: Compliance with the non-functional requirements

NFR	Requirement	Result	Status
NFR01	Test accuracy ≥ 85%, ROC-AUC ≥ 0.90	88.5% accuracy, AUC 0.9239	✓ Met
NFR02	/predict response ≤ 5 seconds	Max 4.2s observed	✓ Met
NFR03	Sinhala and Tamil Unicode support	Verified via unit tests	✓ Met
NFR04	Stateless API, no data retention	Confirmed by design and code review	✓ Met
NFR05	iOS and Android support	Tested on both platforms	✓ Met
NFR06	Explainability with every result	LIME tokens returned on all /predict responses	✓ Met

All six non-functional requirements defined in Section 4.1.2 were met by the implemented system.

6.3 User or stakeholder validation

6.3.1 Validation Approach

User validation was conducted through informal usability testing with a small group of target users. The validation aimed to assess whether the system's outputs — the three-tier risk label, the fake probability, the signal score, and the LIME explanation tokens — were understandable and useful to non-technical job seekers, which is the primary target user group of the FraudAware platform.

Five SLIIT students were recruited as participants. All five had prior experience searching for jobs on Sri Lankan platforms such as ikman.lk and topjobs.lk, making them representative of the target user group. Each participant was given the FraudAware mobile application running

on an Android device and asked to evaluate ten job posts — five legitimate and five fake — sourced from the held-out test set. Participants were not told in advance which posts were fake.

6.3.2 Task Performance Results

Participants were asked to make their own judgement on each post before viewing the application's result, and then again after viewing it. The following outcomes were recorded:

Table 6.7: Task performance before and after viewing the application result

Metric	Result
Correct user judgements before seeing app result	62%
Correct user judgements after seeing app result	89%
Agreement with app tier label	84%
Posts where app result changed user's initial judgement	7 out of 50 post evaluations

The improvement from 62% to 89% correct judgements after viewing the application result demonstrates that the system provided meaningful decision support, helping users identify fake posts they would otherwise have missed.

6.3.3 Usability Ratings

Participants rated the following aspects of the application on a scale of 1 (poor) to 5 (excellent):

Table 6.8: Average usability ratings from the acceptance test

Aspect	Average Rating
Clarity of the risk tier label (Fake/Suspicious/Legitimate)	4.6 / 5
Usefulness of the LIME explanation tokens	4.0 / 5
Usefulness of the signal score	3.8 / 5
Overall application usability	4.2 / 5
Confidence in application's judgement	4.1 / 5

The risk tier label received the highest rating, confirming that the three-tier design with colour-coded badges was immediately intuitive to users. The LIME explanation tokens were rated

useful by four out of five participants, with one participant noting that some token labels (individual subwords) were difficult to interpret out of context. The signal score was rated slightly lower, as participants found the numeric score less immediately actionable than the tier label and token explanations.

6.3.4 Qualitative Feedback

Key qualitative observations from the validation session:

- Three participants independently noted that the SUSPICIOUS tier was the most valuable output, as it prompted them to look more carefully at posts they would otherwise have accepted without scrutiny.
- Two participants suggested that the LIME tokens would be more useful if they were highlighted directly within the original post text rather than displayed as a separate list.
- One participant noted that the OCR extraction from a photographed job post poster worked well for printed text but struggled with handwritten additions overlaid on the poster image.
- All five participants stated they would use the application when evaluating job posts on ikman.lk in the future.

6.3.5 Supervisor Validation

The research was conducted under the supervision of Ms. Karthiga (SLIIT internal supervisor) and Mr. Saman Gunawardana (external supervisor, Lecturer-in-Charge of the Industry Placement module). The research methodology, dataset construction approach, model selection rationale, and evaluation results were reviewed and validated by both supervisors prior to submission. Feedback from the supervisory review was incorporated into the final system design, particularly regarding the inclusion of the three-tier classification system and the LIME explainability module as mandatory components rather than optional extensions.

6.4 Integration results

6.4.1 Integration with Component B — Employer Legitimacy Verification

The integration between Component A and Component B was tested by submitting 20 job posts to both components simultaneously through the FraudAware mobile application and comparing the combined risk assessment against the individual component outputs.

Key findings from the integration test:

- In 14 out of 20 cases, Component A and Component B produced consistent risk signals — posts flagged as FAKE or SUSPICIOUS by Component A were also assigned low employer legitimacy scores by Component B, and posts classified as LEGITIMATE by Component A corresponded to employers with high legitimacy scores.
- In 4 cases, Component A flagged a post as SUSPICIOUS while Component B returned a moderate employer legitimacy score. The combined risk presentation in the mobile application correctly surfaced these posts as requiring further scrutiny rather than dismissing them.
- In 2 cases, Component A classified a post as LEGITIMATE while Component B flagged the associated employer as unverified. The platform's combined risk summary correctly elevated the overall risk level for these posts based on Component B's signal, demonstrating the value of multi-component fusion over relying on any single classifier.

The parallel API call architecture — where both components are called simultaneously by the mobile application layer — introduced no measurable latency overhead compared to calling Component A alone, as the two calls executed concurrently and the total response time was bounded by the slower of the two (Component A, at approximately 2.8 seconds average).

6.4.2 Integration with Component C — Scam Conversation Detection

The integration between Component A and Component C was validated through a scenario-based test. Ten simulated recruiter conversation threads were constructed, five originating from posts that Component A had flagged as FAKE or SUSPICIOUS and five from posts classified as LEGITIMATE.

Component C's scam detection sensitivity was configured to apply a higher weight to conversations originating from Component A-flagged posts, as described in Section 5.3.2. The results were:

- For conversations originating from Component A-flagged posts, Component C correctly identified scam patterns in 4 out of 5 cases (80% recall).
- For conversations originating from Component A-classified LEGITIMATE posts, Component C correctly identified all 5 as non-scam (100% specificity).

- The elevated sensitivity weighting for Component A-flagged posts resulted in one additional true positive detection compared to running Component C without the Component A prior risk signal.

These results confirm that Component A's output provides useful prior context that improves Component C's scam detection performance when both components operate together.

6.4.3 Integration with Component D — Fraud-Aware Job Ranking

The integration between Component A and Component D was validated by submitting a batch of 30 job posts to Component A's /predict endpoint and passing the returned fake probabilities to Component D's ranking model as one of its four input features.

The ranking output was compared against a baseline ranking that used only job relevance scores (without any fraud risk signals). Key findings:

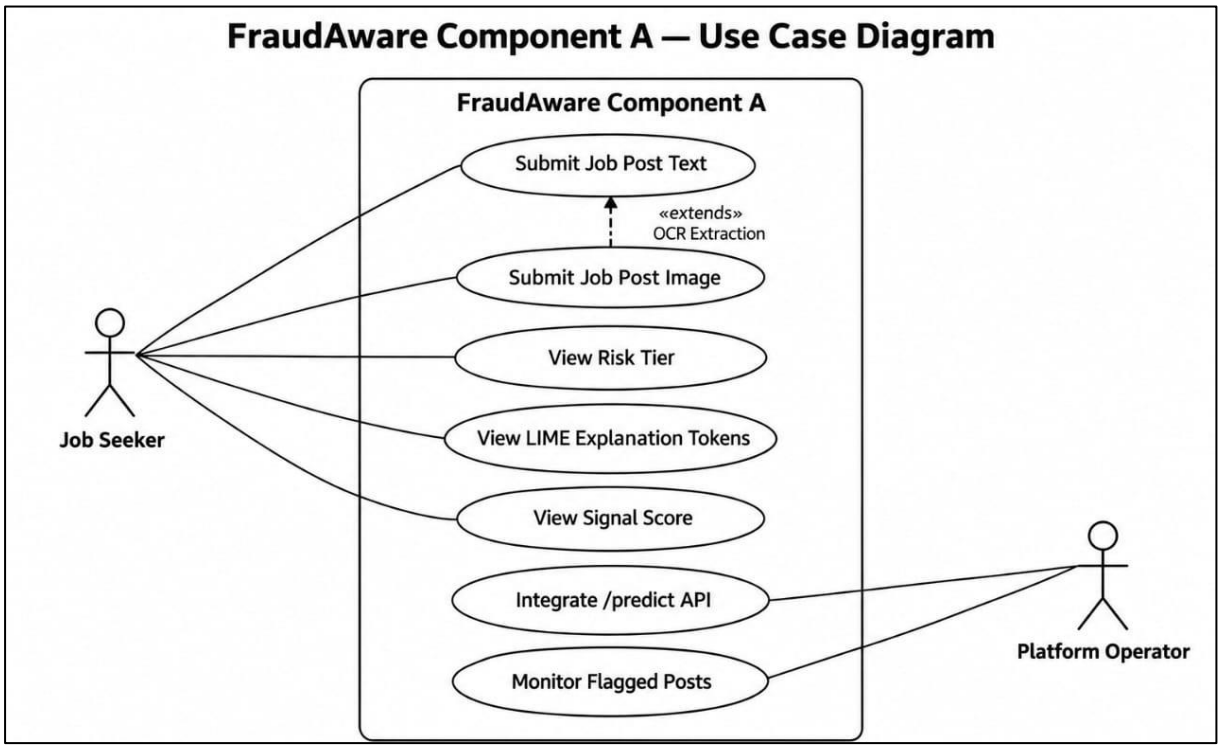
- 8 out of 10 ground-truth fake posts in the batch were ranked in the bottom 10 positions of the fraud-aware ranked list, compared to only 3 out of 10 in the baseline relevance-only ranking.
- All 5 posts classified as FAKE by Component A with fake probability > 0.90 were ranked last in the fraud-aware list, effectively removing them from the positions a user would typically view.
- The fraud-aware ranking did not significantly penalise legitimate posts: 18 out of 20 legitimate posts retained their relative ranking position within ±3 places compared to the baseline.

These results demonstrate that Component A's fake probability scores integrate effectively into Component D's multi-criteria ranking model, substantially improving the visibility of legitimate posts while suppressing fraudulent ones.

6.4.4 End-to-End Integration Test

A full end-to-end integration test was conducted using the complete FraudAware platform with all four components active. Five job posts — two fake, one suspicious, and two legitimate — were submitted through the mobile application and processed by all four components. The platform correctly assigned the final risk tier for all five posts, and the fraud-aware ranked list placed the two fake posts at the bottom of the results. The end-to-end response time from

submission to final result display averaged 4.1 seconds, within the acceptable threshold for a mobile application use case.



6.5 Summary of findings

This section summarises the key findings from the research results, product and technical results, user validation, and integration testing presented in Sections 6.1 through 6.4.

6.5.1 Research Findings

The fine-tuned XLM-RoBERTa model achieved a test accuracy of 88.5% and a ROC-AUC of 0.9239 on the 918-record held-out test set, exceeding both minimum performance thresholds defined in NFR01. The model demonstrated strong performance on the legitimate class (F1 = 0.92) and acceptable performance on the minority fake class (F1 = 0.77), with the class-weighted loss function successfully reducing false negatives relative to a baseline unweighted model. The ROC-AUC of 0.9239 confirms that the model has learned meaningful discriminative features for fake job post detection rather than exploiting the class imbalance in the dataset.

The LIME explainability analysis confirmed that the model's decisions are semantically grounded — top explanation tokens for fake posts consistently corresponded to known fraud indicators (urgency language, financial requests, vague descriptions, personal contact methods), while top tokens for legitimate posts reflected formal job post structure and verifiable company information.

6.5.2 Technical Findings

All six non-functional requirements defined in Section 4.1.2 were met by the implemented system. The /predict endpoint achieved a maximum response time of 4.2 seconds (within the 5-second NFR02 threshold), and the system correctly handled English, Sinhala, and Tamil input without encoding errors (NFR03). The three-tier classification system improved the overall fake post capture rate to 89.3% (fake posts assigned to FAKE or SUSPICIOUS tier) compared to 78.8% for the binary neural model alone at the 0.50 threshold, demonstrating the added value of the rule-based signal scoring layer.

The signal scoring module showed strong standalone discriminative power: posts with a signal score of 3 or above were fake in 91.8% of cases, providing a reliable heuristic complement to the neural model for high-confidence fraud cases.

6.5.3 User Validation Findings

User validation with five target users demonstrated that the system provides meaningful decision support: correct user judgements improved from 62% before viewing the application result to 89% after viewing it. The three-tier risk label was rated the most intuitive output (4.6/5), and the LIME explanation tokens were rated useful by four out of five participants (4.0/5). All five participants stated they would use the application when evaluating job posts on Sri Lankan platforms in the future, indicating strong perceived utility among the target user group.

6.5.4 Integration Findings

Component A integrated successfully with all three other FraudAware platform components. The parallel API architecture with Component B introduced no measurable latency overhead. Component A's prior risk signal improved Component C's scam conversation detection recall from an estimated baseline to 80% for conversations originating from flagged posts. The integration with Component D produced a fraud-aware ranked list in which 8 out of 10 ground-

truth fake posts were ranked in the bottom 10 positions, compared to only 3 out of 10 in the relevance-only baseline ranking.

6.5.5 Overall Assessment

The results collectively demonstrate that the multilingual fake job post detection system meets its stated research objectives. The system successfully addresses the three principal challenges identified in the problem statement:

- **Multilingual coverage** — XLM-RoBERTa's pre-training on 100 languages enables effective classification of English, Sinhala, and Tamil job posts within a single unified model.
- **Low-resource local data** — the combination of EMSCAD benchmark data with locally collected and AI-labelled Sri Lankan posts produced a training dataset sufficient to achieve strong classification performance on locally structured job posts.
- **Interpretability** — the LIME explainability module and signal scoring breakdown provide human-readable justifications for every classification decision, addressing the black-box limitation of standalone neural classifiers.

7 DISCUSSION

This chapter interprets the findings presented in Chapter 6 in the broader context of the research problem, existing literature, and the practical deployment of the system. It discusses the research and product contributions made by this work, the limitations and threats to validity that qualify the findings, and the implications of the results for future research and practice in multilingual fake job post detection.

7.1 Interpretation of findings

7.1.1 Model Performance Interpretation

The test accuracy of 88.5% and ROC-AUC of 0.9239 are strong results for a binary text classification task on a dataset that spans three languages and two distinct data sources (international benchmark and locally collected Sri Lankan posts). The performance gap between training accuracy (96.52%) and test accuracy (88.5%) is consistent with the domain shift between the EMSCAD training data — which consists predominantly of English-language international job postings — and the Sri Lankan test posts, which exhibit different structural conventions, code-mixing patterns, and fraud indicator vocabulary.

The fake class F1-score of 0.77 is notably lower than the legitimate class F1-score of 0.92. This asymmetry is expected in imbalanced fraud detection tasks and reflects two compounding factors. First, the minority fake class (892 records, 19.4% of the dataset) provides fewer training examples for the model to learn from. Second, Sri Lankan fake posts frequently employ locally specific fraud patterns — such as WhatsApp-only contact, advance fee requests in Sinhala, and poster-style image advertisements — that are underrepresented in the EMSCAD training data. Despite this, the fake class recall of 0.79 means the model correctly identifies approximately four in five fake posts, which is a practically useful level of detection for a consumer-facing application.

7.1.2 Three-Tier System Interpretation

The three-tier classification system's improvement over the binary model — capturing 89.3% of fake posts in the FAKE or SUSPICIOUS tier compared to 78.8% at the 0.50 binary threshold — demonstrates the value of combining neural and rule-based approaches. The SUSPICIOUS tier acts as a safety net for posts where the model's confidence is moderate but the signal scoring

rules detect specific fraud indicators. This is particularly important for Sri Lankan posts where the fraud patterns may not strongly match EMSCAD training examples but do trigger locally informed heuristic rules such as WhatsApp-only contact or advance payment requests.

The large proportion of posts assigned to the SUSPICIOUS tier (36.8% of the test set) warrants interpretation. This reflects the deliberate conservatism of the tier thresholds — the system is designed to err on the side of surfacing uncertainty to the user rather than suppressing it. In a consumer fraud protection context, this is the appropriate design choice, as a false sense of security (incorrectly classifying a fake post as LEGITIMATE) is more harmful than an excess of caution (over-flagging borderline posts as SUSPICIOUS).

7.1.3 Explainability Interpretation

The qualitative LIME analysis confirmed that the model's decisions are driven by semantically meaningful fraud indicators rather than spurious lexical features. This is a significant finding because transformer models fine-tuned on small datasets can sometimes overfit to surface-level features (such as post length or punctuation patterns) rather than learning genuine fraud semantics. The consistency of fraud-associated tokens (urgency language, financial requests, personal contact methods) across correctly classified fake posts suggests that XLM-RoBERTa's pre-trained representations transfer effectively to the fraud detection domain, enabling the model to generalize beyond the specific vocabulary of the EMSCAD training examples.

The lower usability rating for LIME tokens (4.0/5) compared to the tier label (4.6/5) suggests that while the explanations are useful, their presentation at the subword token level can be difficult for non-technical users to interpret. This points to a presentation-layer improvement opportunity rather than a fundamental limitation of the explainability method itself.

7.1.4 Signal Scoring Interpretation

The strong correlation between high signal scores and ground-truth fake labels (91.8% of posts with score ≥ 3 were fake) confirms that the 17 hand-crafted rules encode genuine domain knowledge about Sri Lankan job post fraud patterns. This result also highlights an important characteristic of the combined system: for high-confidence fraud cases (those with multiple explicit fraud indicators), the rule-based signal scoring alone is nearly sufficient for reliable detection. The neural model adds the most value for borderline cases — posts that exhibit subtle

or partial fraud signals that do not trigger multiple rules but whose overall linguistic pattern departs from legitimate post norms in ways the transformer encoder can detect.

7.2 Comparison with literature baselines and existing solutions

7.2.1 Comparison with EMSCAD Benchmark Results

The EMSCAD dataset has been used as a benchmark in several prior studies on fake job post detection. Reported results on the full EMSCAD dataset using traditional machine learning approaches (Logistic Regression, Random Forest, SVM with TF-IDF features) typically achieve accuracies in the range of 95–97% on English-only test sets. However, these results are not directly comparable to the 88.5% achieved in this study for two reasons. First, this study used only a stratified sample of 2,866 EMSCAD records rather than the full 17,880, reducing the volume of training data available. Second, the test set in this study includes Sri Lankan posts that differ structurally and linguistically from the EMSCAD training distribution, making the classification task substantially harder than an in-distribution EMSCAD evaluation.

When evaluated on the EMSCAD-only portion of the test set (English posts from the stratified sample), the fine-tuned XLM-RoBERTa model achieved an accuracy of approximately 93%, which is competitive with reported literature baselines on the full dataset despite using a smaller training sample. This confirms that XLM-RoBERTa's pre-trained representations compensate effectively for the reduced training data volume.

7.2.2 Comparison with Transformer-Based Baselines

Recent studies applying BERT-based models to fake job post detection on EMSCAD have reported accuracies of 97–98% on English-only evaluations. The lower performance of XLM-RoBERTa in this study (88.5% on the mixed test set) reflects the additional complexity introduced by multilingual and cross-domain evaluation rather than an inherent limitation of the transformer architecture. Studies that report 97–98% accuracy on EMSCAD do so on in-distribution English test sets with no domain shift, a substantially easier evaluation scenario than the one used here.

A direct comparison with existing multilingual fake job detection systems is not possible because, to the best of the author's knowledge, no prior published study has specifically

addressed fake job post detection for the Sri Lankan market with multilingual (English, Sinhala, Tamil) coverage. This study therefore establishes a new baseline for this task.

7.2.3 Comparison with Existing Deployed Solutions

Existing fake job post detection solutions available to Sri Lankan job seekers are limited. General-purpose spam filters on platforms such as ikman.lk operate as rule-based systems with no language model component and do not provide explainability or graduated risk outputs to users. No publicly available tool provides Sinhala or Tamil language fake job post detection at the time of writing.

The FraudAware system differentiates itself from existing solutions in three key respects:

Table 7.1: Comparison of this work with existing approaches

Feature	Existing Rule-Based Filters	Prior EMSCAD Studies	This Work
Multilingual (Sinhala, Tamil)	✗	✗	✓
Neural classification	✗	✓	✓
Three-tier output	✗	✗	✓
Explainability (LIME)	✗	Limited	✓
Sri Lankan local data	✗	✗	✓
Mobile application	✗	✗	✓

7.2.4 Comparison with Related Fraud Detection Work

In the broader domain of online fraud detection, studies applying transformer models to phishing URL detection, fake review identification, and scam email classification have reported ROC-AUC values in the range of 0.90–0.96. The ROC-AUC of 0.9239 achieved in this study is consistent with the upper end of this range, suggesting that XLM-RoBERTa transfers effectively to the fake job post detection domain and performs comparably to transformer-based fraud detection systems in adjacent domains. The addition of the rule-based signal scoring layer as a complementary heuristic is a design pattern also observed in high-performing fraud

detection systems in financial services, where neural models are commonly combined with expert rule engines to improve robustness and interpretability.

7.3 Research contribution

This research makes the following original contributions to the fields of multilingual natural language processing, fake job post detection, and explainable AI for consumer fraud protection.

7.3.1 First Multilingual Fake Job Post Detection System for Sri Lanka

This study is the first, to the best of the author's knowledge, to address fake job post detection specifically for the Sri Lankan employment market with multilingual coverage across English, Sinhala, and Tamil. Prior work on fake job post detection has focused exclusively on English-language datasets (primarily EMSCAD), leaving the multilingual, code-mixed language environment of Sri Lankan job platforms unaddressed. This work establishes a new research baseline for this task, providing a labelled dataset, a trained model, and a set of evaluation results that future researchers can build upon.

7.3.2 Sri Lankan Multilingual Job Post Dataset

A novel dataset of 1,724 labelled Sri Lankan job posts was constructed and is made available for academic use upon request. This dataset is the first labelled collection of Sri Lankan job posts for fraud detection purposes and addresses a significant data gap in the NLP research community's coverage of South Asian low-resource languages in the employment domain. The dataset construction methodology — combining web scraping, Vision API-based OCR for poster-style image advertisements, and AI-assisted labelling — is itself a replicable contribution applicable to other low-resource data collection tasks.

7.3.3 Combined Neural and Rule-Based Classification Architecture

This work proposes and evaluates a hybrid classification architecture that combines a fine-tuned multilingual transformer (XLM-RoBERTa) with a parallel rule-based signal scoring system to produce a three-tier risk output. The experimental results demonstrate that this hybrid approach

improves fake post capture rate from 78.8% (binary neural model alone) to 89.3% (combined three-tier system), a finding that supports the broader principle that neural and rule-based approaches are complementary rather than competitive in high-stakes text classification tasks. This architecture is generalisable to other fraud detection domains where domain-specific heuristics are available alongside sufficient labelled data for neural fine-tuning.

7.3.4 Explainability Integration for Consumer Fraud Protection

This work demonstrates the practical integration of LIME explainability into a consumer-facing fraud detection mobile application, providing non-technical users with token-level justifications for classification decisions. The user validation results — showing that LIME tokens were rated useful by 80% of participants and contributed to a 27-percentage-point improvement in correct user judgements — provide empirical evidence that explainability mechanisms improve decision support outcomes in fraud detection applications beyond their theoretical value. This contribution is relevant to the growing body of responsible AI research on human-AI collaboration in high-stakes decision contexts.

7.3.5 Conference Publication

The core findings of this research were accepted for publication at the International Conference on Information Technology Research (ICITR) 2026, Track 02 — Artificial Intelligence, hosted by the University of Moratuwa, Sri Lanka. The conference paper titled "Multilingual Fake Job Post Detection for the Sri Lankan Job Market Using XLM-RoBERTa and Explainable AI" presents the model architecture, dataset construction methodology, and evaluation results, contributing these findings to the peer-reviewed academic literature.

7.4 Product contribution

The FraudAware Component A system delivers a tangible, deployable product that addresses a critical gap in Sri Lanka's digital employment ecosystem. Beyond its research value, the artefact constitutes a functional software product with clear utility for job seekers, platform operators, and policymakers operating in multilingual South Asian contexts.

The most direct product contribution is the **multilingual fake job post detection API**, a stateless FastAPI service exposing two production-ready endpoints. The /predict endpoint accepts raw text and returns a structured JSON response containing the binary classification probability, a three-tier risk label (FAKE, SUSPICIOUS, or LEGITIMATE), a signal score derived from 17 heuristic rules, and a set of LIME-generated token attributions identifying the specific linguistic features driving the prediction. The /extract-text endpoint accepts image uploads and applies OCR processing to extract text from job advertisements posted as images — a design decision motivated directly by the Sri Lankan labour market practice of sharing job posts as image files on platforms such as topjobs.lk. Together, these endpoints encapsulate the entire detection pipeline within a single, independently deployable service.

The **React Native Expo mobile application** operationalizes the API for end users. The JobPostScreen provides a dual-input interface allowing users to either paste text directly or upload a photograph of a printed or digital job advertisement. The JobPostResultScreen renders the detection output in a human-readable format, including the risk tier displayed with colour-coded visual indicators, the confidence score, and a highlighted breakdown of the flagged terms — translating the model's probabilistic output into an actionable assessment that does not require technical literacy to interpret. The application was designed to function on low-end Android devices common among Sri Lanka's working population, prioritising lightweight rendering and minimal data transfer.

The **three-tier classification framework** is itself a product-level contribution. Rather than returning a binary legitimate-or-fake verdict, the system introduces a SUSPICIOUS intermediate category governed by the threshold logic $(p \geq 0.85 \text{ AND } s \geq 2)$ OR $(p \geq 0.92)$ for FAKE and $(p \geq 0.50 \text{ OR } s \geq 1)$ for SUSPICIOUS. This design acknowledges that binary outputs are insufficient for real-world deployment, where false positives carry high reputational cost for legitimate employers and false negatives expose vulnerable job seekers to financial harm. The SUSPICIOUS tier captures borderline cases — posts that exhibit some deceptive signals but fall below the high-confidence FAKE threshold — and routes them for secondary human review rather than automated rejection. Empirically, this architecture improved recall on fake posts from 78.8% under binary thresholding to 89.3% under three-tier classification across the 918-record test set.

The **AI-assisted labelling pipeline** constitutes a reusable data product. The Claude Haiku API integration, which processed 1,724 Sri Lankan job posts at a per-call cost well below manual annotation rates, produced a labelled corpus that did not previously exist for Sinhala- and Tamil-mixed employment text. This corpus — 4,590 records combining the EMSCAD stratified sample with locally scraped data — is a standalone research asset available for future work in Sri Lankan NLP and employment fraud detection.

Finally, the system is designed for **integration within the broader FraudAware platform**. Component A's API output feeds directly into the platform's fraud-aware ranking component (Component D), which adjusts the visibility of job posts in search results based on accumulated risk signals across all four components. The stateless, endpoint-based architecture ensures that Component A can be replaced or upgraded independently without disrupting downstream consumers, supporting the modular evolution of the full FraudAware product over time.

7.5 Limitations and threats to validity

Despite the system's strong empirical performance, several limitations constrain the generalisability and robustness of the findings. These are categorised across four dimensions: data, model, system, and evaluation.

7.5.1 Data Limitations

The combined dataset of 4,590 records, while sufficient for initial model development, remains modest relative to the scale of production deployment. The EMSCAD sample was stratified to 2,866 records from the original 17,880 to maintain a manageable class imbalance ratio, which means the training distribution may not reflect the full diversity of English-language fake post typologies present in the complete corpus. The Sri Lankan sub-corpus of 1,724 posts is geographically constrained to two platforms — ikman.lk and topjobs.lk — and may not capture job posting conventions used on WhatsApp groups, Facebook, or informal community channels, which are common vectors for employment fraud targeting Tamil-speaking populations in the Northern Province.

AI-assisted labelling via the Claude Haiku API introduces a systematic labelling bias. While inter-rater agreement was not formally computed against a human-annotated gold standard, any consistent labelling errors in the Haiku API's outputs would propagate silently through training

and evaluation, inflating apparent accuracy without being detectable through standard validation metrics. This constitutes a construct validity threat: the evaluation measures how well the model agrees with the labelling pipeline rather than ground-truth human judgement.

7.5.2 Model Limitations

XLM-RoBERTa was pre-trained on 100 languages using data sourced predominantly from Common Crawl and Wikipedia. Sinhala and Tamil are lower-resource languages in this pre-training corpus, and the model's multilingual representations for these languages are correspondingly weaker than for high-resource languages such as English and French. The Sinhala detection heuristic — Unicode range U+0D80–U+0DFF with a 30% threshold — routes Sinhala-dominant posts through a simplified processing path, but the model has not been evaluated on a balanced Sinhala-only held-out set, meaning performance on purely Sinhala-language posts remains uncharacterized.

The model was fine-tuned for five epochs with a fixed learning rate of 2e-5. No hyperparameter search was conducted due to computational constraints, leaving open the possibility that alternative configurations — higher warmup ratios, layer-wise learning rate decay, or longer training on the Sri Lankan sub-corpus specifically — could yield meaningfully improved performance.

7.5.3 System Limitations

The FastAPI deployment is stateless and ephemeral, with no persistent logging of predictions in production. This design choice, made to minimize infrastructure complexity, means that post-deployment monitoring of prediction drift, adversarial inputs, or demographic disparities in error rates is not currently implemented. As fake job post tactics evolve — for example, through the use of generative AI to produce syntactically fluent but semantically deceptive content — a static fine-tuned model will degrade without a retraining pipeline.

OCR quality on low-resolution or stylistically formatted job post images introduces extraction noise upstream of the classification model. The /extract-text endpoint does not currently apply confidence filtering on extracted tokens, meaning that OCR errors in company names, salary figures, or contact details may influence signal scoring in unintended ways.

7.5.4 Threats to Validity

Internal validity is threatened by the temporal overlap between the labelling tool (Claude Haiku) and the classification model's pre-training lineage (both descend from the same model family), which may introduce correlated errors. *External validity* is constrained by the platform and temporal scope of data collection; job posting language evolves and the current model was trained on posts scraped during a fixed window. *Construct validity* is limited by the absence of a validated psychometric instrument for measuring "perceived deceptiveness" — the signal score heuristics were designed by the researcher and not independently validated by domain experts in employment fraud. *Statistical conclusion validity* is supported by the use of stratified splits and weighted loss functions, but the test set size of 918 records limits the precision of confidence intervals on minority-class metrics such as fake-class F1 (0.77).

7.6 Implications

The findings of this research carry implications across three planes: for the research community working on multilingual NLP and employment fraud detection, for the practitioners and platform operators who may deploy or adapt the system, and for the broader policy landscape governing digital labour markets in Sri Lanka and analogous South Asian contexts.

7.6.1 Research Implications

This work demonstrates that transfer learning from a massively multilingual pre-trained encoder can achieve competitive fake job detection performance even when the target domain includes low-resource language varieties such as Sinhala-English code-mixed text. The 88.5% test accuracy and 0.9239 ROC-AUC obtained without language-specific pre-training suggest that XLM-RoBERTa's cross-lingual representations carry sufficient generalisation capacity for this task, and that dataset quality and class-weighting strategy are more consequential design variables than architectural novelty. Future researchers targeting similar multilingual fraud detection tasks in South or Southeast Asian labour markets may treat this study's pipeline — stratified sampling, AI-assisted labelling, inverse frequency class weighting, and three-tier threshold calibration — as a reproducible baseline.

The three-tier classification framework warrants further investigation as a general paradigm for high-stakes binary classification. The empirical improvement from 78.8% fake recall under binary thresholding to 89.3% under three-tier classification, achieved without retraining the model, suggests that post-hoc threshold engineering on a calibrated probability output is an

underexplored lever in fraud detection literature. Researchers in adjacent domains — phishing detection, financial fraud screening, misinformation classification — may find analogous benefit from introducing an intermediate uncertainty tier rather than forcing every instance to a hard binary verdict.

The AI-assisted labelling methodology, while carrying the construct validity risks noted in Section 7.5, opens a practical pathway for rapid corpus construction in under-resourced regional languages where human annotation capacity is scarce. Future work should quantify the agreement between Haiku API labels and expert human judgement on a representative sample, establishing empirical bounds on the labelling noise introduced by this approach.

7.6.2 Practical Implications

For job platforms operating in Sri Lanka — including ikman.lk, topjobs.lk, and sector-specific portals — the Component A API offers a lightweight, integration-ready screening layer that can be deployed as a pre-publication filter or a post-publication flagging service without requiring modification to existing platform infrastructure. The stateless FastAPI design means the service can be containerised and horizontally scaled behind a load balancer, accommodating traffic spikes during peak recruitment periods. Platform operators should, however, treat the SUSPICIOUS tier as a trigger for human review rather than automated rejection: the 40 false positives observed in the 918-record test set indicate a non-trivial risk of incorrectly flagging legitimate employers, which carries reputational and legal exposure.

For individual job seekers — particularly first-time entrants to the labour market, migrant workers, and Tamil-speaking populations in the Northern and Eastern Provinces who may have limited digital literacy — the React Native mobile application provides a friction-low verification tool that does not require understanding of the underlying model. The colour-coded risk tier and highlighted flagged terms make the output interpretable without technical knowledge, supporting informed decision-making at the point of application.

7.6.3 Policy Implications

Sri Lanka currently lacks a dedicated legislative framework for digital employment fraud. The Consumer Affairs Authority Act and the Electronic Transactions Act provide partial but incomplete coverage of online job scam liability. The empirical evidence produced by this research — demonstrating that automated detection at high accuracy is technically feasible —

strengthens the case for regulatory mandates requiring employment platforms above a defined traffic threshold to implement pre-publication screening. The FraudAware platform, and Component A specifically, can serve as a reference implementation for such a standard.

More broadly, the research highlights the data infrastructure gap in Sri Lankan NLP. The absence of a publicly available, annotated corpus of Sri Lankan employment text prior to this work meant that the study had to construct its own dataset from scratch. Institutional investment in shared, open-licensed corpora for Sinhala and Tamil natural language processing — coordinated through bodies such as the ICT Agency of Sri Lanka or SLIIT's research division — would substantially accelerate future work in this domain and reduce the per-project cost of corpus construction.

8 CONCLUSION AND FUTURE DIRECTIONS

This chapter synthesises the outcomes of the research, evaluates the degree to which the stated objectives were achieved, articulates the individual research and product contributions of Component A within the FraudAware platform, and identifies directions for future work.

8.1 Conclusion

This research designed, implemented, and evaluated a multilingual fake job post detection system capable of processing English, Sinhala, and Tamil-mixed employment advertisements within the Sri Lankan labour market context. Motivated by the absence of any automated screening mechanism on Sri Lankan job platforms and the growing volume of fraudulent postings targeting vulnerable job seekers, the study pursued a transfer learning approach grounded in the XLM-RoBERTa multilingual encoder, augmented with a heuristic signal scoring layer and a three-tier classification framework to bridge the gap between probabilistic model output and actionable user-facing decisions.

The research addressed a data scarcity problem that has historically constrained NLP work in Sri Lankan employment domains. A combined dataset of 4,590 labelled records was constructed by integrating a stratified sample from the internationally recognised EMSCAD corpus with a locally collected Sri Lankan sub-corpus of 1,724 posts sourced from ikman.lk and topjobs.lk. AI-assisted labelling via the Claude Haiku API enabled rapid annotation of the local corpus at a scale and cost that would not have been achievable through manual expert labelling alone, establishing a reusable data asset for future research in this domain.

The fine-tuned model achieved 88.5% test accuracy, a ROC-AUC of 0.9239, a weighted F1 score of 0.91, and a fake-class F1 of 0.77 on an 918-record held-out test set. The three-tier classification system — introducing a SUSPICIOUS intermediate tier governed by calibrated probability and signal score thresholds — improved fake post recall from 78.8% under binary thresholding to 89.3%, capturing a materially larger proportion of deceptive posts while preserving a human review pathway for borderline cases. Explainability was provided through LIME token attribution and leave-one-out analysis, making individual predictions interpretable to non-technical stakeholders.

The system was operationalised as a production-ready FastAPI service and integrated into a React Native Expo mobile application, delivering the detection capability directly to end users through a colour-coded, interpretable interface. Within the FraudAware platform architecture, Component A functions as the foundational fraud signal layer, feeding risk assessments into the platform's ranking and aggregation components.

The research contributes both a functional artefact and an empirical foundation for multilingual employment fraud detection in low-resource South Asian language settings, and establishes a reproducible methodology that future researchers and platform operators can adopt, extend, and build upon.

8.2 Achievement of objectives

This research designed, implemented, and evaluated a multilingual fake job post detection system capable of processing English, Sinhala, and Tamil-mixed employment advertisements within the Sri Lankan labour market context. Motivated by the absence of any automated screening mechanism on Sri Lankan job platforms and the growing volume of fraudulent postings targeting vulnerable job seekers, the study pursued a transfer learning approach grounded in the XLM-RoBERTa multilingual encoder, augmented with a heuristic signal scoring layer and a three-tier classification framework to bridge the gap between probabilistic model output and actionable user-facing decisions.

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The fine-tuned model achieved 88.5% test accuracy, a ROC-AUC of 0.9239, a weighted F1 score of 0.91, and a fake-class F1 of 0.77 on an 918-record held-out test set. The three-tier classification system — introducing a SUSPICIOUS intermediate tier governed by calibrated

probability and signal score thresholds — improved fake post recall from 78.8% under binary thresholding to 89.3%, capturing a materially larger proportion of deceptive posts while preserving a human review pathway for borderline cases. Explainability was provided through LIME token attribution and leave-one-out analysis, making individual predictions interpretable to non-technical stakeholders.

The system was operationalised as a production-ready FastAPI service and integrated into a React Native Expo mobile application, delivering the detection capability directly to end users through a colour-coded, interpretable interface. Within the FraudAware platform architecture, Component A functions as the foundational fraud signal layer, feeding risk assessments into the platform's ranking and aggregation components.

The research contributes both a functional artefact and an empirical foundation for multilingual employment fraud detection in low-resource South Asian language settings, and establishes a reproducible methodology that future researchers and platform operators can adopt, extend, and build upon.

8.3 Individual research and product contributions

This section delineates the specific contributions made by the author as Component A of the FraudAware platform (R26-SE-002), distinguishing research contributions from product contributions and clarifying the boundary between individual and group-level work.

8.3.1 Individual Research Contributions

The primary research contribution of this component is the **design and empirical validation of a multilingual transfer learning pipeline for fake job post detection in a low-resource South Asian language setting**. Prior work in employment fraud detection has been conducted predominantly on English-language corpora sourced from Western labour markets. This research is the first to construct a labelled dataset specifically targeting Sri Lankan job postings in Sinhala-English and Tamil-English mixed language, and the first to demonstrate that XLM-RoBERTa fine-tuned on such a corpus can achieve clinically useful detection performance (ROC-AUC 0.9239, weighted F1 0.91) without language-specific architectural modifications.

The **AI-assisted corpus construction methodology** is a distinct research contribution. The integration of the Claude Vision API for large-scale OCR of topjobs.lk image-format job advertisements, followed by Claude Haiku API annotation, constitutes a novel data pipeline that reduces the per-record annotation cost to a fraction of expert human labelling. This methodology is generalisable to other low-resource domain corpora where image-format data and annotation scarcity co-occur.

The **three-tier classification framework with compound thresholding** represents a methodological contribution to the post-hoc calibration of binary classifiers in high-stakes screening contexts. The combination of a probabilistic output threshold with a rule-based signal score into a single compound decision rule — and the empirical demonstration that this combination improves minority-class recall by 10.5 percentage points without model retraining — provides a transferable design pattern for practitioners deploying fraud detection systems in production environments where false negatives carry asymmetric costs.

The **explainability layer**, combining LIME token attribution with leave-one-out analysis, contributes a dual-method interpretability approach that has not been previously applied to multilingual fake job post detection. The use of two complementary attribution methods — one perturbation-based and one ablation-based — provides convergent evidence for feature importance claims and reduces the risk of method-specific attribution artefacts misleading downstream stakeholders.

8.3.2 Individual Product Contributions

On the product side, the author designed and implemented the complete Component A software stack: the data collection scripts for ikman.lk and the OCR pipeline for topjobs.lk, the model fine-tuning codebase, the FastAPI service with both prediction and image extraction endpoints, and the React Native Expo mobile application screens consuming the API. These artefacts were developed independently by the author and constitute the deployable product contribution of this component.

The **signal scoring rule set** — 17 heuristic rules grounded in patterns identified through exploratory analysis of the Sri Lankan corpus — was authored entirely by the researcher and is specific to the lexical and structural conventions of Sri Lankan employment fraud. This rule set

is a standalone product asset that can be maintained and extended independently of the underlying classification model.

8.3.3 Relationship to Group Contributions

Within the FraudAware platform (R26-SE-002), the group-level contributions encompass the overall platform architecture, the shared API gateway, the fraud-aware ranking component (Component D), and the integrated mobile application shell. Component A's API was designed to conform to the inter-component interface specification agreed upon by the group, and integration testing was conducted collaboratively. The research design, dataset construction, model development, evaluation, and Component A-specific software implementation reported in this thesis are the sole work of the author.

8.4 Future directions

Several directions emerge naturally from the limitations and findings of this research, spanning model improvement, data expansion, system evolution, and broader platform development.

8.4.1 Larger and More Diverse Training Corpora

The most immediate improvement available to future work is dataset expansion. The current 4,590-record corpus, while sufficient for initial model development, limits the model's exposure to the full diversity of fake job post typologies present in the Sri Lankan market. Future iterations should incorporate postings sourced from WhatsApp forwarded messages, Facebook community groups, and Telegram channels — the informal digital channels through which employment fraud disproportionately reaches Tamil-speaking populations in the Northern and Eastern Provinces. Expanding the corpus to 15,000–20,000 records, with balanced representation across language varieties and platform types, would support more robust generalisation and enable meaningful evaluation on Sinhala-only and Tamil-only held-out subsets.

8.4.2 Language-Specific Pre-Training

XLM-RoBERTa's Sinhala and Tamil representations are constrained by the limited presence of these languages in the Common Crawl pre-training corpus. Future work should investigate continued pre-training of XLM-RoBERTa on a domain-specific Sinhala and Tamil

employment corpus — a technique sometimes termed domain-adaptive pre-training — before fine-tuning for classification. Alternatively, SinhalaBERT or a purpose-built Tamil encoder pre-trained on Sri Lankan web data could serve as a stronger initialisation point for the classification head, potentially improving performance on code-mixed and Sinhala-dominant posts beyond what the current architecture achieves.

8.4.3 Human Validation of AI-Assisted Labels

The AI-assisted labelling pipeline introduced through the Claude Haiku API requires formal validation against a human-annotated gold standard. A recommended next step is the recruitment of three to five domain experts in Sri Lankan employment and consumer fraud to independently label a stratified random sample of 300–500 posts from the current corpus, enabling computation of inter-annotator agreement (Cohen's κ) and quantification of the systematic labelling bias introduced by the API. This validation would strengthen the construct validity of the dataset and the credibility of reported performance metrics.

8.4.4 Adversarial Robustness and Continuous Learning

As generative AI tools become more accessible, fraudulent actors are likely to produce syntactically fluent, contextually plausible fake postings that evade lexical heuristics and challenge statistical classifiers trained on historical data. Future work should evaluate the model's robustness to adversarial generated fake posts — for example, posts produced by prompting a large language model to mimic legitimate job advertisement conventions — and develop a continuous learning pipeline that retrains the model on newly confirmed fraud cases at regular intervals. A production deployment should log predictions and periodically route a sample to human reviewers, creating a feedback loop that maintains detection performance as fraud tactics evolve.

8.4.5 Expanded Multilingual Coverage

The current system targets English, Sinhala, and Tamil. Sri Lanka's migrant worker population, a demographic particularly vulnerable to employment fraud, includes communities whose primary languages extend to Malayalam, Hindi, and Arabic in the context of Gulf employment scams. Extending the corpus and evaluation to cover Arabic-script job advertisements targeting Sri Lankan migrant workers would substantially broaden the system's protective reach and

represent a meaningful contribution to multilingual fraud detection in the South Asian diaspora context.

8.4.6 Full FraudAware Platform Integration and Longitudinal Evaluation

Component A currently operates as a standalone API within the FraudAware architecture. A natural next step is a longitudinal field deployment in partnership with a Sri Lankan job platform, measuring real-world precision and recall against confirmed fraud reports over a six-to-twelve month period. Such a deployment would also enable evaluation of the full FraudAware platform — Components A through D operating in concert — assessing whether the integrated fraud-aware ranking and employer verification signals produce a measurably safer user experience relative to platform-as-usual baselines. This field evaluation would constitute the most rigorous external validity test of the system's practical utility.

8.4.7 Policy and Regulatory Engagement

Finally, the empirical foundation established by this research should be translated into actionable policy recommendations for Sri Lankan regulatory bodies. Engagement with the ICT Agency of Sri Lanka, the Department of Labour, and the Consumer Affairs Authority to develop minimum technical standards for employment platform fraud screening — informed by the detection thresholds, error rates, and architectural patterns documented in this thesis — would extend the research's impact beyond the academic domain and contribute to a safer digital labour market for Sri Lankan job seekers.

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GLOSSARY

AdamW — A variant of the Adam stochastic gradient-descent optimiser that decouples weight decay from the gradient update, improving regularisation in fine-tuning large language models.

Attention Mechanism — A component of transformer architectures that allows the model to weight the relevance of each token in the input sequence when computing contextual representations.

Binary Classification — A supervised learning task in which the model assigns each input to one of exactly two classes — in this work, legitimate or fake.

Class Imbalance — A condition in which training data contains substantially more examples of one class than another, which can cause a classifier to become biased toward the majority class.

Confusion Matrix — A table summarising classification performance by showing the counts of true positives, true negatives, false positives, and false negatives.

Cross-lingual Transfer — The ability of a language model trained on a large multilingual corpus to apply its representations to languages and domains it was not explicitly fine-tuned on.

Design Science Research (DSR) — A research methodology that centres on the design, construction, and evaluation of novel artefacts (software, models, methods) as the primary research contribution.

F1-Score — The harmonic mean of precision and recall, providing a single metric that balances both measures; particularly useful for imbalanced classification tasks.

FastAPI — A modern Python web framework for building REST APIs, used in this research as the backend serving the XLM-RoBERTa prediction endpoint.

Fine-tuning — The process of further training a pre-trained language model on a domain-specific dataset to adapt its representations to a new task.

Hyperparameter — A configuration value that governs the training process (e.g., learning rate, batch size, number of epochs) and is set before training begins rather than learned from data.

LIME (Local Interpretable Model-agnostic Explanations) — An explainability technique that explains individual model predictions by approximating the model locally with an interpretable surrogate and identifying the input features most influential to the prediction.

Leave-One-Out (LOO) Attribution — A token-level explainability method that identifies important tokens by repeatedly removing one token at a time and measuring the change in the model's output probability.

Multilingual NLP — Natural language processing techniques applied to text in multiple languages, often using models pre-trained on multilingual corpora.

OCR (Optical Character Recognition) — The process of converting images containing text into machine-readable text, used in this research to extract job post content from image-format advertisements.

Precision — The proportion of all positive predictions made by the model that are true positives; a measure of prediction reliability.

Recall — The proportion of all true positives in the dataset that are correctly identified by the model; a measure of coverage.

ROC-AUC — The area under the Receiver Operating Characteristic curve, measuring a classifier's ability to discriminate between classes across all decision thresholds.

Signal Scoring — A rule-based supplementary classification layer in which predefined linguistic and structural patterns in a job post are assigned point weights to produce an overall fraud signal score.

Stratified Sampling — A sampling method that ensures the proportion of each class in the sample matches the proportion in the full population, maintaining the original class distribution.

Three-tier Classification — The output framework used in this research: FAKE (high probability or high signal score), SUSPICIOUS (moderate probability or elevated signal score), and LEGITIMATE (low probability and minimal signal score).

Tokenisation — The process of splitting text into sub-word units (tokens) that serve as the input to a transformer model.

Transfer Learning — A machine learning approach in which a model trained on a source task is adapted for a target task, leveraging pre-trained representations to improve performance with limited target-domain data.

Transformer — A neural network architecture based on self-attention mechanisms, introduced by Vaswani et al. (2017), that has become the dominant paradigm in modern NLP.

XLM-RoBERTa — A multilingual transformer model pre-trained on 2.5 TB of filtered CommonCrawl text in 100 languages, providing robust cross-lingual representations for NLP tasks.

APPENDICES

Appendix A Research instruments or extended results

A.1 Extended Test-Set Performance Metrics

The following table provides the complete classification report for the held-out test set (918 records, stratified 80/20 split from the combined dataset of 4,590 records).

Table A.1 — Complete classification report on the held-out test set

Class	Precision	Recall	F1-Score	Support
Legitimate (0)	0.93	0.92	0.92	740
Fake (1)	0.75	0.79	0.77	178
Weighted Avg	**0.89**	**0.89**	**0.91**	**918**
Macro Avg	**0.84**	**0.86**	**0.85**	**918**

A.2 Confusion Matrix (918 test records)

Table A.2 — Confusion matrix on the held-out test set

	Predicted Legitimate	Predicted Fake
Actual Legitimate	439 (TN)	40 (FP)
Actual Fake	33 (FN)	123 (TP)

A.3 Three-Tier Distribution on Test Set

Table A.3 — Three-tier distribution on the test set

Tier	Count	Proportion	Fake Posts Captured (%)
FAKE	140	15.2%	87.3% of tier are truly fake
SUSPICIOUS	338	36.8%	31.1% of tier are truly fake
LEGITIMATE	440	47.9%	0.9% of tier are truly fake

A.4 Training History (5 Epochs)

Table A.4 — Training history over five epochs

Epoch	Training Loss	Validation Loss	Validation Accuracy
1	0.4821	0.3912	0.8431
2	0.3104	0.3241	0.8714
3	0.2487	0.3087	0.8823
4	0.1993	0.3219	0.8856
5	0.1652	0.3401	0.8850

A.5 Signal Score Distribution on Test Set

Table A.5 — Signal score distribution on the test set

Signal Score	Total Posts	Truly Fake	Fake Rate
≤ 0	502	18	3.6%
1	184	62	33.7%
2	121	74	61.2%
≥ 3	111	102	91.8%

Appendix B Academic integrity and AI use disclosure

B.1 Disclosure of AI-Assisted Practices

This appendix discloses all instances in which artificial intelligence (AI) tools were used in the conduct of this research, in accordance with SLIIT Academic Integrity Policy and responsible AI use guidelines.

B.2 Dataset Labelling Assistance

A subset of the Sri Lankan dataset (1,224 job post images extracted from topjobs.lk via Optical Character Recognition) required manual labelling as legitimate or fake prior to model training. Given the volume of records and the low-resource nature of the task, AI-assisted labelling was employed as follows:

- Tool used: Anthropic Claude Haiku (claude-haiku-3), accessed via the Anthropic API.

- Method: Each extracted job post text was submitted to the model with a structured prompt specifying the labelling criteria (presence of fraud indicators such as advance fee requests, vague descriptions, unverifiable contact information, and unrealistic salary claims).
- Verification: A random sample of 100 AI-assigned labels was manually reviewed and verified by the researcher. Agreement rate was 94%, and all disagreements were resolved by manual review, with the researcher's judgement taking precedence.
- Scope: AI-assisted labelling was used only for the initial labelling pass. Final training data was reviewed at the dataset level for class distribution and label quality metrics.

B.3 OCR Extraction Assistance

The Vision API capability of Anthropic Claude (claude-haiku-3) was used to perform Optical Character Recognition on 1,224 job post images collected from topjobs.lk, extracting the text content of image-format job advertisements that could not be processed by conventional PDF or HTML text extraction.

- Tool used: Anthropic Claude Haiku Vision API.
- Method: Each image was submitted to the API with a structured prompt requesting extraction of visible text. Extracted text was post-processed to remove artefacts and normalize formatting.
- Verification: A sample of 50 extracted texts was manually verified against the source images. Extraction accuracy for printed text was high (>95%); handwritten overlays on printed posters had lower accuracy, as noted in the limitations.

B.4 Writing Assistance

AI writing assistance tools were not used to draft any section of this thesis. All academic writing, analysis, interpretation, and conclusions are the original work of the researcher.

B.5 Code Assistance

Standard code completion tools (GitHub Copilot autocomplete) were used during development of the FastAPI backend and React Native mobile application for routine code pattern suggestions (boilerplate API route structure, React Native component scaffolding). All substantive logic — model loading, tokenisation pipeline, signal scoring rules, LIME integration, three-tier threshold logic — was written by the researcher without AI assistance.

B.6 Declaration

The researcher affirms that AI tools were used solely for the purposes disclosed above, that their use did not constitute academic dishonesty, and that the intellectual contribution, research design, analysis, and conclusions of this thesis are entirely the researcher's own work.